Azimuth-Aware Subspace Classifier for Few-Shot Class-Incremental SAR ATR

Yan Zhao, *Graduate Student Member, IEEE*, Lingjun Zhao, Siqian Zhang, Kefeng Ji, *Member, IEEE*, Gangyao Kuang, and Li Liu, *Senior Member, IEEE*

***Abstract*— With the rapid acquisition of high-resolution synthetic aperture radar (SAR) images, new categories are continually observed with few-shot instances in openly noncooperative scenarios. Powering an SAR automatic target recognition (SAR ATR) system with an ability of few-shot class-incremental learning (FSCIL) is nontrivial. Observing the pronounced azimuth-dependence and part-sparsity of targets in SAR images, an azimuth-aware subspace classifier (AASC) on the Grassmannian manifold is proposed to tackle the FSCIL of SAR ATR stably and accurately. In the AASC, losses covering both semantic and manifold facets, which include semantic margin separation (SMS), deep subspace separation (DSS), and structure less forgetting (SLF), are designed to strike both the intrinsic model's stability and plasticity dilemma and domain-specific challenges. For plasticity, the novel-to-old semantic margins are enlarged by the SMS loss for knowledge transferring while avoiding inappropriate adaptions. The DSS loss derived from the Grassmannian geometry aims to regularize class subspaces orthogonality. For stability, semantic drifts of target spatial and global structures are punished by the SLF loss. As the periodicity and volatility of target azimuth-aware patterns, an azimuth-aware exemplar selection (AES) strategy is designed to select representative and complementary exemplars. In experiments, the advantages of the subspace classifier and the designed losses and strategies are deeply verified. Comprehensive experiments on three FSCIL scenarios derived from both airborne and spaceborne datasets, including the moving and stationary target acquisition and recognition (MSTAR), the SAR-AIRcraft-1.0, and self-collected datasets, show that our method significantly outperforms various task-specific benchmarks, verifying its effectiveness for the FSCIL in real-SAR ATR scenarios.**

***Index Terms*— Deep learning (DL), few-shot class-incremental learning (FSCIL), Grassmannian manifold, subspace classification, synthetic aperture radar (SAR) automatic target recognition (ATR).**

I. Introduction

SYNTHETIC aperture radar (SAR) has superior and stable imaging capabilities in nearly all weather and illumination conditions, which plays an indispensable role in a wide variety of civil and military applications, such as military surveillance, surveillance [1], [2], [3], border security [4], [5], and so on. As a fundamental and challenging task of SAR imagery interpretation, SAR automatic target recognition (SAR ATR) has received longstanding and intensive attention for several decades. In particular, the SAR ATR has reached unprecedented prosperity advanced by deep learning (DL) techniques.

Nowadays, with the rapid progress of SAR imaging techniques, the amount of high-resolution SAR images is growing exponentially, leading to a sharp increase in the observation of new categories of targets. More practically, constrained by openly dynamic environments, e.g., noncooperative military battlefield, platform resources, e.g., satellites, unmanned aerial vehicles (UAVs), vehicles, and security rules, e.g., data privacy and confidentiality, few-shot new-class targets can be observed continually while storing the old ones for knowledge replay may be low-efficient and even forbidden. Therefore, it is urgently essential for current DL-based SAR ATR methods to facilitate the ability of few-shot class-incremental learning (FSCIL) to satisfy the practical scenarios. Needless to say, with the powerful feature representing ability, the DL-based classifier learned on entire samples can achieve overwhelming performance compared with many incremental-learning (IL) solutions and even suppress human beings in traditional computer vision, e.g., incredible accuracy for recognizing a thousand different classes of objects in natural images [6], and SAR ATR fields [7], [8], [9], [10], [11]. However, directly transferring these approaches to the FSCIL of SAR ATR is suboptimal and incompatible due to the following shortcomings.

1) **Catastrophic Forgetting:** In real-world scenarios, new categories are usually observed with stream format as the amount of SAR images increases. A canonical DL-based method must be retrained repeatedly on all collections to acquire considerable performance in all classes. However, caused by the huge sizes, e.g., tens of kilometers in imaging width, of high-resolution SAR images and the access privacy, it is intractable and expensive to store entire data of seen targets to dynamically update models in practical scenarios, especially for edge devices, such as robots, satellites, and UAVs. Moreover, data privacy and security may also limit the acquisition of all training data, even target-related chips. Consequently, the DL-based methods can easily erase the former knowledge when learning the new, leading to irreversible performance degradation.
2) **Inaccurate Representation:** Due to the particularity of SAR imaging mechanisms, acquiring large-scale diversified targets of interest in SAR images is still expensive

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Yan Zhao, Lingjun Zhao, Siqian Zhang, Kefeng Ji, and Gangyao Kuang are with the State Key Laboratory of Complex Electromagnetic Environment Effects on Electronics and Information System, National University of Defense Technology, Changsha 410073, China (e-mail: zy18@nudt.edu.cn; nudtzlj@163.com; zhangsiqian@nudt.edu.cn; jikefeng@nudt.edu.cn; kuangyeats@hotmail.com).

Li Liu is with the College of Electronic Science, National University of Defense Technology, Changsha 410073, China (e-mail: dreamliu2010@gmail.com).

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compared with those in natural images and even impossible in most noncooperative situations. Besides, precisely annotating large-scale training samples to train a DL learner is labor-intensive and time-costing, resulting from the strong requirement of professional domain knowledge for labeling. Therefore, the informative cues from few-shot samples are incomplete with high uncertainty, which inevitably causes a biased empirical risk for the data-oriented DL-based solutions to learn novel class knowledge.

Striking the above-mentioned problems resorts to addressing the model's **stability and plasticity dilemma** [12], with the stability indicating the model's ability to remember previous knowledge and the plasticity representing the model's ability to learn new patterns continually. In particular, addressing the FSCIL of SAR ATR is unique and intractable due to the following domain challenges.

1) **Insufficient Target-Discerning Information:** In contrast to rich color, illumination, and appearance cues contained in natural images, targets in SAR images are merely represented as discretely backscattering points reflecting their structural patterns (see pseudoimages of Fig. 1 for an example), which are confusing and insufficient. Therefore, it is essential to fully utilize target structural cues provided by samples at limited azimuth angles for stable identification without sacrificing generalization.
2) **Seriously Interclass Similarity and Intraclass Variation:** Due to the similar components, structures, and layouts among different classes, elements of interclass targets in SAR images are easily confused at specific azimuth angles. Conversely, obvious discrepancies might exist within the interclasses at greatly different imaging conditions, such as target azimuths and radar depression angles. Thus, models designed for the FSCIL of SAR ATR should pay attention to targets at special azimuth angles to make intraclass features compact and interclass ones separated.
3) **Complex Background Interference:** Intensive background clutters widely exist around the targets in SAR images due to the complexity and nonhomogeneity of surroundings, seriously causing much interference to DL-based methods for extracting and discriminating unseen samples accurately by using few-shot samples. Thus, an elegant classification strategy with special attention to target details and robustness to circumstances is highly demanded.

In response to the general and domain-specific problems, an azimuth-aware subspace classifier (AASC) is proposed to solve the FSCIL of SAR ATR on the Grassmannian manifold, benefiting from its compatibility and discriminability to targets and background robustness to clutters. Notably, knowledge of targets is formed as class-aware subspaces by using their structural patterns with strong azimuth priors preextracted on a deep cosine space. Besides, different from directly estimating and adopting real azimuth information, which is inaccurate, complicated, and impractical in real scenarios, the information is primarily mapped into a deep cosine space, benefiting from the natural relationship between azimuth dependence of target structures and the phase sensitivity of cosine criterion. Consequently, unknown targets can be identified by a point-to-space metric on a Grassmannian manifold. In the AASC, at the base learning stage, where large-scale training data is provided to imitate offline learning in cooperative situations, a CNN-based feature extractor is trained by a cosine cross-entropy (CE) loss to establish a semantic hypersphere to capture target azimuth-aware structural details. Then, subspaces of base classes and the corresponding exemplars selected by the AES are saved for later evaluation. At the IL stages, where only few-shot samples are provided to mimic an online learning process, losses covering both projected and semantic levels, which contain semantic margin separation (SMS), deep subspace separation (DSS), and structure less forgetting (SLF), are leveraged to mitigate both the intrinsic dilemma and the domain-specific challenges. For the model's plasticity, the SMS loss is designed to enlarge discrepancies of targets around the decision boundaries to achieve clear separations between new and old categories. The DSS loss, which considers the projection metric derived from the principle angles of the Grassmannian manifold, is introduced to make subspaces as orthogonal as possible for knowledge separation. For the model's stability, knowledge of old classes is preserved by constraining semantic drifts of spatial and global target structures mapped by the former fixed and the current trainable models in the SLF loss. In particular, considering the prominent periodicity and volatility of target backscattering features as azimuth changes, an azimuth-aware exemplar selection (AES) strategy is proposed to select complementary and representative samples for efficiently preserving knowledge of old classes.

In experiments, the advantages of the subspace classifier and the proposed losses and strategies for solving the FSCIL problem of SAR ATR are explored. Extensive experiments on three FSCIL scenarios separately derived from both airborne and spaceborne datasets, including the moving and stationary target acquisition and recognition (MSTAR), the SAR-AIRcraft-1.0, and a self-collected one, show that our method outperforms various task-specific advanced benchmarks by a large margin, verify its effectiveness for surviving in openly dynamic SAR ATR scenarios.

Overall, this article has the following threefold contributions.

1) We unlock the FSCIL of SAR ATR and propose an AASC framework to first learn and identify targets on the deep Grassmannian manifold, considering its advantages of target discriminability and background robustness.
2) We design losses, including the SMS, the DSS, and the SLF, and an AES strategy to cope with the intrinsic stability and plasticity dilemma concerning domain characteristics and challenges of SAR ATR.
3) Three FSCIL scenarios derived from the MSTAR, the SAR-AIRcraft-1.0, and self-collected datasets are constructed, and our method achieves the state-of-the-art (SOTA) performance compared with numerous task-specific advanced benchmarks.

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In what follows, we describe our solution in detail. In Section II, the related works of SAR target classification, few-shot learning (FSL), and the FSCIL are concisely provided. Afterward, we detail the problem definition of the FSCIL, the motivations, and our method in Section III. Experimental setups, results, and analysis are provided in Section IV. Finally, we give the conclusions in Section VI.

II. Related Works

A. SAR Target Classification

The problems of SAR target classification and the manners to deal with them have been extensively studied for several decades. The approaches could be broadly taxonomized into template-based, model-based, and machine-learning-based paradigms. In the template-based scheme [13], [14], [15], targets are classified by comparing them to preestablished class templates based on low-level handcrafted cues, such as grayscale, length, edge, and region moment. Although simple and intuitive the features are, acquiring them precisely and stably is challenging due to the diverse target postures and complex backgrounds in SAR images. In the model-based methods [16], [17], [18], [19], target electromagnetic scattering patterns estimated by parameterized models are used for similarity comparison. Although strongly explainable the methods are, precisely representing target scattering patterns is intractable due to the complex imaging mechanisms and the simplified models. The machine-learning-based, especially the DL-based methods [20], [21], [22], [23], could stably represent target features at high-dimensional spaces, which leads to accurate predictions. However, most approaches are data-hungry and can merely perform well on the predefined classes.

This article still aims to solve the problem of SAR target classification and proposes a DL-based solution, specifically aiming to learn continually from few-shot samples of new classes without losing performance on the old.

B. Few-Shot Learning

FSL aims to alleviate label-efficient issues and has been extensively researched in the past. Methods for the problem fall broadly into meta-learning and nonmeta-learning schemes [24]. The former scheme [25], [26], [27] focuses on making DL-based models obtain an ability to adapt to new classes by using few-shot samples after a large number of episodic learning circles. Then, real-class samples could be identified at a meta-testing stage. Although new concepts can be quickly captured by these task-agnostic methods, their training and inferring pipelines are complicated and inflexible. In contrast, nonmeta-learning-based methods [28], [29], [30], [31] usually make predictions based on the sophisticated distance criteria between features of testing samples and predefined classes, which is simple and flexible. Considering limited samples collected in most SAR ATR situations, numerous works [9], [32], [33], [34], [35], [36], [37], [38] have devoted to bringing few-shot techniques into this field, and considerable improvements have been achieved. However, they mainly consider the current classes but ignore the performance of the old.

In our method, knowledge of novel classes is also provided by few-shot samples. In particular, its IL ability is especially considered.

C. Few-Shot Class-Incremental Learning

FSCIL [39] is a practical yet very challenging problem and aims to teach a model to incrementally learn novel knowledge without forgetting the old. As a special scope of IL, most of the solutions for the FSCIL are directly borrowed from the classical IL. Among these, knowledge distillation has been broadly used, benefiting from its simplicity and effectiveness. In this paradigm, methods [40], [41], [42] overcome the problem by penalizing the drifts of multilevel outputs, e.g., logits, features, and relational, produced by the former and the current models. Besides, considering the insufficient knowledge provided by the few-shot samples of novel classes, meta-learning-based methods [43], [44], [45] derived from the FSL are also introduced to solve the problem. Here, methods are trained to acquire the ability to continually learn from few-shot novel samples after learning on numerous episodic tasks at the meta-training stage. Then, the unseen classes could be recognized at the meta-testing stage. Following the paradigm, Wang, et al. proposed a hierarchical embedding and incremental evolutionary network (HEIEN) network [46], in which an adaptive class-IL (ACIL) module was meta-trained to adaptively adjust weights of all seen classes.

The proposed method follows the knowledge-distillation-based paradigm and first solves the FSCIL of SAR ATR in the Grassmannian manifold, considering domain knowledge and challenges to satisfy the practical demands.

III. Methodology

We detail the proposed method in this part. The problem setup of the FSCIL is defined in Section III-A. The motivations are given by Section III-B. The overall architecture of our method is depicted in Section III-C. The subspace construction and classification strategies are illustrated in Section III-D with losses described in Section III-E. The proposed exemplar selection strategy is detailed in Section III-F.

A. Problem Setup

FSCIL aims to enable classifiers to continually learn new knowledge from few-shot samples without forgetting the previously learned ones [39]. To mimic a practical learning scenario, the FSCIL usually contains one base and several incremental steps, a.k.a., sessions. In the base session, large-scale training data per class, serving as precollected knowledge in cooperative situations, are provided. In incremental sessions, new-class samples continually appear to form online learning in noncooperative conditions. At each session, a classifier should adapt quickly by using new-class samples and even a few exemplars of old classes. At inference, all seen categories until the current session should be well-identified.

Formally, supposing that we have a sequence of labeled datasets $\mathcal{D}^1, \mathcal{D}^2, \ldots, \mathcal{D}^t$, where $\mathcal{D}^t = \{x_i^t, y_i^t\}_{i=1}^N$. x_i^t and y_i^t are each training sample and its category. $\mathcal{C}^t$ is the class set of $\mathcal{D}^t$. Specifically, for $\forall\ j, k \in t,\ j \neq k, \mathcal{C}^j \cap \mathcal{C}^k = \varnothing$ holds. Usually, the base session dataset $\mathcal{D}^1$ contains large-scale training

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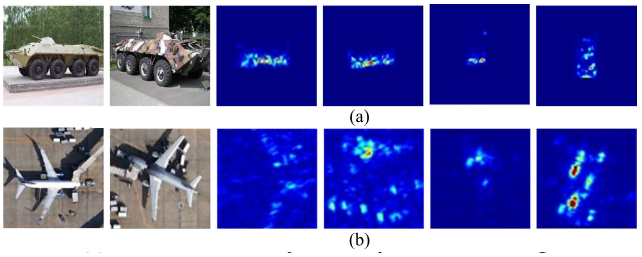


Fig. 1. Illustration of the azimuth dependence and part sparsity properties of targets in SAR images. Targets imaged by both airborne and spaceborne SAR images are provided in (a) and (b) figures separately. (a) BTR70 vehicle from the MSTAR dataset. (b) A320 aircraft from the SAR-AIRcraft-1.0 dataset. Best viewed in pseudocolor for target SAR images.

samples per class. For each incremental session $t, t > 1$, the dataset $\mathcal{D}^t$ contains several new classes, which are composed as an N-way K-shot task with small N and K. At session i, a model can be optimized on the $\mathcal{D}^t$ and possibly a small set of old-class samples E^{t-1}. At inference, the model is expected to discriminate all seen classes $\mathcal{C} = \cup_{i=1}^{t} \mathcal{C}^i$.

B. Motivations

In SAR imagery, target-discerning structures are mainly represented as diversified distributions of backscattering points, exhibiting distinct properties of **azimuth dependence** and **part sparsity**.

1) **Azimuth Dependence:** In contrast to constantly distinct characteristics of targets in optical images, those in SAR images change dynamically with the pronounced property of azimuth dependence raised by the specificity of SAR imaging mechanisms and diverse target postures. In Fig. 1(a) and (b), two types of targets, i.e., a BTR70 vehicle and an A320 aircraft, imaged by airborne and spaceborne SAR platforms are provided along with their optical images. From the last four columns of Fig. 1, targets at different azimuth angles (postures) exhibit greatly different backscattering patterns.
2) **Part Sparsity:** Apart from the earlier, high-energy parts of targets are distributed sparsely in both airborne and spaceborne SAR images. In contrast to rich target details, such as textures, shapes, and colors, in optical images, only specific strong backscattering structures can be distinctly reflected in SAR images. In particular, these structures are sometimes constant or complementary with the change of target postures. For example, the turret of the BTR70 and the engines of the A320 in SAR images can exhibit constantly salient scattering energy at most postures. However, the wings of the A320 can be merely observed at special azimuth angles.

As the above-mentioned two properties and the scarcity of target-discerning cues provided by few-shot samples, identifying targets accurately and stably calls for fully utilizing their diversified patterns remained at a wide range of azimuth angles. In contrast to modeling target knowledge as classical 1-D vectors, such as prototypes, nodes of FC layers, and making an inference based on a one-to-one matching, e.g., the Euclidean and cosine distances, the merits of representing and identifying targets by subspace classifiers on a Grassmannian manifold are threefold.

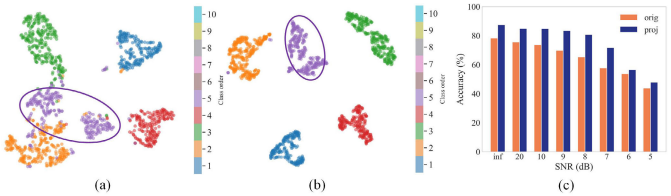


Fig. 2. Illustration of the advantages of subspaces. (a) tSNE of five-class features at the original spaces. (b) tSNE of five-class features projected by class-aware subspaces. (c) Classification accuracy based on the original and the projected features as image quality degrades.

1) *Intraclass Compactness:* Compared with formulating knowledge of classes as blurry and tightly coupled classifier nodes, a class subspace constructed by a set of targets at various azimuth angles can explicitly contextualize class-aware structural cues, making features of the same classes compact, and those of different ones loosely coupled, significantly improving the model's stability to preserve the learned knowledge and compatibility to incorporate the new. As a toy example, the t-distributed stochastic neighbor embedding (tSNE) of five-class (four bases + one novel) features at the original spaces, and those mapped by class-aware subspaces are shown in Fig. 2(a) and (b), respectively. The class-aware projected features are more compact than those at the original spaces, especially for the incremental class (see the purple scatters for an example), suggesting the potential of representing discerning targets compactly by subspace classifiers.
2) *Interclass Separation:* Due to the existence of similar configurations and structures, backscattering patterns of different categories may be highly confused at specific imaging conditions, which easily leads to indistinct class boundaries at original feature spaces. However, a subspace composed of target principle cues can fully capture target-discerning azimuth-aware information and maintain clear separations among different classes, implicitly improving the model's discriminability. In comparison with overlapped feature points of different classes [Fig. 2(a)], the correspondingly projected features [Fig. 2(b)] are with distinct separations among different classes.
3) *Background Robustness:* The nonhomogeneous surroundings easily cause serious interference to target identification. Besides, due to the existence of similar configurations and structures, features of new classes may also cause serious class-level interference with the old ones during the model's continual adaption in the FSCIL of SAR ATR. In contrast to classifying targets based on confused and redundant target patterns at original semantic spaces, target-discerning azimuth-aware information can be fully captured and contextualized within subspaces, improving the model's robustness to backgrounds and discriminability to different targets. As image quality [single noise rate (SNR)] degrades, the classification accuracy based on the original and the projected features is shown in Fig. 2(c). It can be seen that the performance obtained based on the projected

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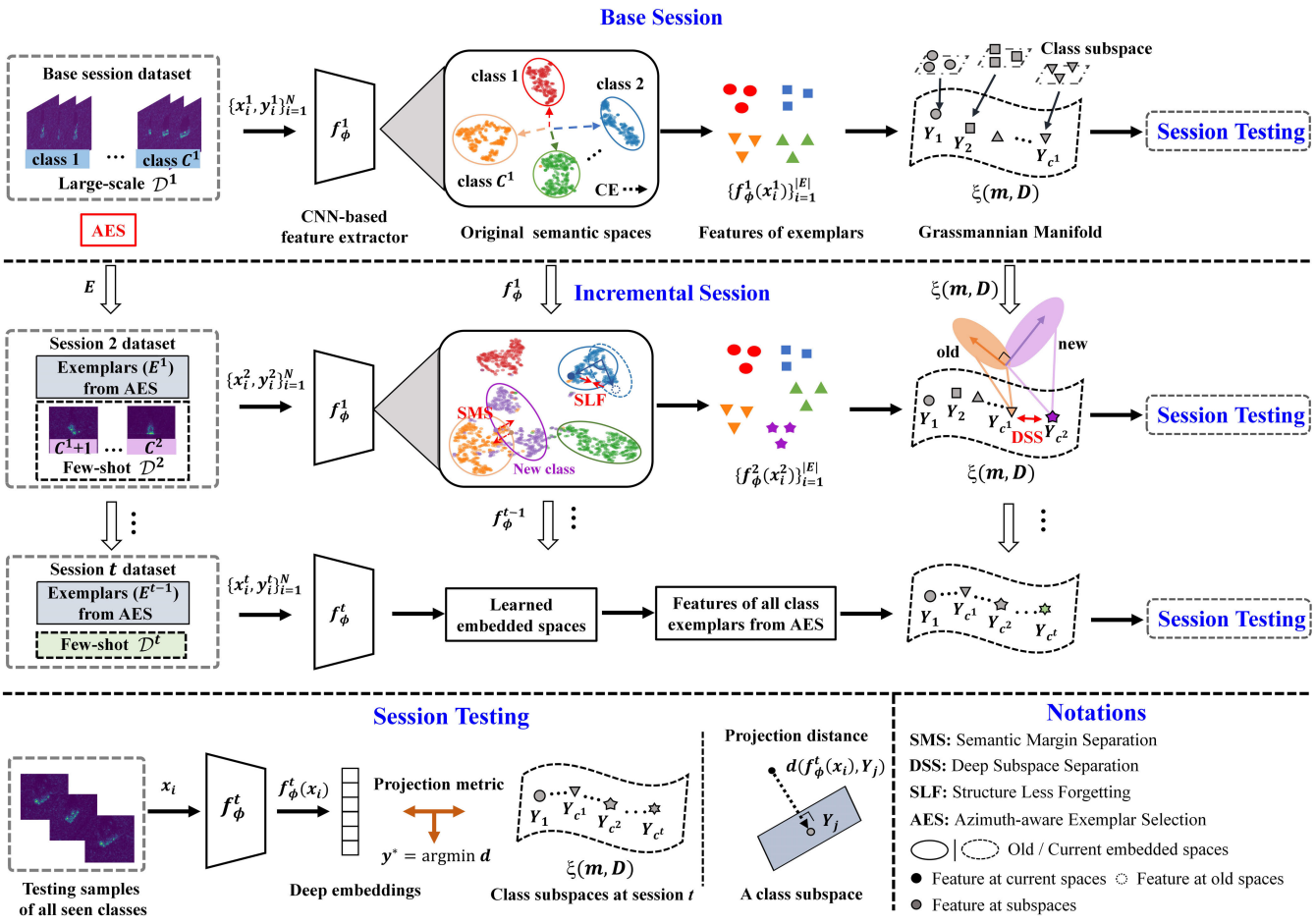


Fig. 3. Flowchart of the AASC. A well-spanned CNN-based feature extractor is first constructed using $\mathcal{D}^1$ at the base session. New-class samples and the saved exemplars are combined at each incremental session for optimization. At the testing stages, classification is done on the Grassmannian manifold based on the projection distance between deep features of testing samples and subspaces of all seen classes. Three losses, which include SMS, DSS, and SLF, and an exemplar selection strategy, AES, are proposed (marked by the red font) to address the model's stability and plasticity dilemma.

features is more stable and superior to that based on the original.

In the FSCIL of SAR ATR, maintaining compact and stable structures for intraclass knowledge and well-separated boundaries among different classes is crucial to mitigate the stability and plasticity dilemma. Considering the two domain-specific properties and the aforementioned advantages of subspace classifiers at the manifold, constructing a framework for FSCIL of SAR ATR based on subspaces is highly preferred to solve the problem.

C. Overall Framework

The overall framework of our method is shown in Fig. 3 and detailed in this part.

At the base session ($t = 1$), a CNN-based feature extractor parameterized as f_ϕ^1 is trained by a cosine CE [47] loss to classify base classes and extract general characteristics of novel classes at later incremental sessions. Considering the natural relationship between azimuth dependence of target backscattering structures in SAR images and the phase sensitivity of cosine criterion, target-discerning information with implicit azimuth-aware priors can be effectively captured by the f_ϕ^1 for further evaluation. Then, exemplars of base classes E^1 are selected by the proposed AES strategy for knowledge replay. Meanwhile, we construct class-specific subspaces, i.e., $Y_1, Y_2, \ldots, Y_{c^1}$, corresponding to classes from 1 to c^1, on the Grassmannian manifold $\xi(m, D)$ by using features from E^1 for evaluation.

At the incremental sessions ($t \geq 1$), taking the second ($t = 2$) as an example, the few-shot novel class dataset $\mathcal{D}^2$ with classes ranging from $c^1 + 1$ to c^2 and old-class exemplars E^1 are combined as the training data of the second session. To balance the model's plasticity for new knowledge adaption and the stability for old knowledge preservation, losses covering both the original semantic space and the Grassmannian manifold, which include the SMS, the SLF, and the deep-aware subspace separation (DSS) (the red fonts in Fig. 3), are designed. For plasticity, the SMS loss is adopted to enlarge separations among current classes based on the triplet cosine distances. In particular, considering the limited knowledge from the saved exemplars and new class samples, only semantic margins of novel-to-old class samples at near azimuth angles are enlarged to avoid inappropriate knowledge transferring. In the DSS loss, distances between different subspaces at the $\xi(m, D)$, i.e., $Y_1, Y_2, \ldots, Y_{c^2}$, are measured and pushed to be orthogonal based on the projection metric

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derived from cosine-based principle angles. For knowledge preservation, we adopt the knowledge distillation in the SLF loss for simplicity and efficiency. The high-level semantic drifts of both spatial and global target features produced by the former fixed and the current trainable models are regularized to stabilize both detailed and global target structures. After session learning, exemplars with both representativeness and complementarity are selected by the AES for efficient knowledge rehearsal and subspace construction under the guidance of feature-level energy and azimuth-aware cosine correlations. We formulate the training procedures as Algorithm 1. The model's plasticity ($\mathcal{L}_{\text{DSS}}$, $\mathcal{L}_{\text{SMS}}$) and the stability ($\mathcal{L}_{\text{SLF}}$) are balanced by the γ, which is defined as $((|\mathcal{C}^{\text{old}}|/|\mathcal{C}^t|))^{1/2}$. Here, the denominator and numerator are the numbers of old and new classes at session t, respectively.

At the session testing stage, as demonstrated in the bottom-left subfigure of Fig. 3, the label y^* of a testing sample x_i from all seen categories is figured out by finding the closest distance $d(f_\phi^t(x_i), Y_j)$ of the deep embedding of x_i to all its projections onto different subspaces $Y = \cup_{j=1}^{c^t} Y_j$.

Algorithm 1 Training Procedures of Our Method

Input: A CNN-based feature extractor f_ϕ, a sequence of labeled data $\mathcal{D}^1$,$\mathcal{D}^2$,...,$\mathcal{D}^t$

Output: The trained f_ϕ^t, subspaces $Y = \cup_{j=1}^{c^t} Y_j$ at the Grassmannian maifold.

Step 1: Base session learning on $\mathcal{D}^1$

initialize a proxy classifier parameterized as W

for *epoch* $\leftarrow 1$ to N epochs **do**

Compute $\mathcal{L}_{CE} = -y_i \log(p(x_i))$, where $p(x_i) = \sigma(W^\top f_\phi^1(x_i))$ on $\mathcal{D}^1$, σ is a Softmax function

Update f_ϕ^1 and W until convergence

end for

Saved exemplars E^1 by the AES from $\mathcal{D}^1$ and f_ϕ^1, construct subspaces $Y = \cup_{j=1}^{c^1} Y_j$ from E^1

Step 2: Incremental session learning on $\mathcal{D}^t, t > 1$

Construct session dataset by combing E^{t-1} and $\mathcal{D}^t$

for *epoch* $\leftarrow 1$ to M **do** epochs

Compute $\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{SMS}} + \mathcal{L}_{\text{DSS}} + \gamma \mathcal{L}_{\text{SLF}}$

Update f_ϕ^t until convergence

end for

Saved exemplars E^t by the AES from $\mathcal{D}^t$ and f_ϕ^t, re-build class-aware subspaces $\cup_{j=1}^{c^t} Y_j$ from exemplars of all seen class E^t

D. Subspace Construction and Classification

In this section, the definition and characteristics of the Grassmannian manifold are briefly introduced, followed by elaborations on the subspace construction and classification.

1) Introduction of the Grassmannian Manifold: The Grassmannian manifold is a set of m-dimensional linear subspaces of $\mathbb{R}^D$ and can be notated as $\xi(m, D)$. The $\xi(m, D)$ is a $m(D-m)$ compact Riemannian manifold. An element of $\xi(m, D)$ can be represented by an orthonormal matrix Y of size $D \times m$, satisfying $Y^\top Y = I_m$, where I_m is an identity matrix with the size of $m \times m$. In the view of the Grassmannian manifold, a subspace span(Y) defined in the Riemannian manifold of $\mathbb{R}^D$ can be seen as a point in $\xi(m, D)$. The two points Y_1 and Y_2 are the same only and only if span(Y_1) = span(Y_2). That is $Y_1 R_1 = Y_2 R_2$ for some $R_1, R_2 \in \mathcal{O}(m)$, where $\mathcal{O}(m)$ is a group of orthonormal matrix with size of m by m.

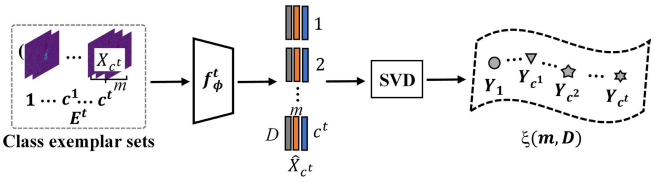


Fig. 4. Illustration of subspace construction at session t.

The distances between two subspaces span(Y_1) and span(Y_2) can be regraded as distances of points at $\xi(m, D)$ measured by the principle angles $\boldsymbol{\theta} = [\theta_1, \theta_2, \ldots, \theta_m]$, where $0 \le \theta_1 \le \theta_2 \le \cdots \le \theta_m \le \pi/2$. The angles are defined as follows and can be obtained by the matrix decomposition of $Y^\top Y$

$$\begin{aligned} \cos\theta_k = \max_{\boldsymbol{u_k} \in \text{span}(Y_1)} \max_{\boldsymbol{v_k} \in \text{span}(Y_2)} & \boldsymbol{u_k}^\top \boldsymbol{v_k} \\ \text{s.t. } & \boldsymbol{u_k}^\top \boldsymbol{u_k} = 1, \quad \boldsymbol{v_k}^\top \boldsymbol{v_k} = 1 \\ & \boldsymbol{u_k}^\top \boldsymbol{u_i} = 0, \quad \boldsymbol{v_k}^\top \boldsymbol{v_i} = 0, \quad (i = 1, \ldots, k-1). \end{aligned} \tag{1}$$

Notably, since the pronounced properties of azimuth dependence and part sparsity of targets in SAR images as well as serious background interference, target features are first mapped by the f_ϕ into a semantic hypersphere space to capture their azimuth-aware structures cues. Furthermore, representing the mapped class-aware features as points at $\xi(m, D)$ could preserve more rich and intrinsic structural links within few-shot samples than vanilla semantic vectors, implicitly improving the model's ability of knowledge preservation and adaption.

2) Subspace Construction: To construct subspace of class c^t, as diagramed in Fig. 4, a matrix $\hat{X}_{c^t} = [f_\phi^t(1), f_\phi^t(2), \ldots, f_\phi^t(m)] \in R^{m \times D}$ is first formed based on the semantic features of class exemplars extracted by f_ϕ^t, where m and D refer to the number of exemplars of class c^t and the dimension of the flattened features from f_ϕ^t, respectively. Then, we employ the singular value decomposition (SVD) to decompose $\hat{X}_{c^t}$ into $Y \sum V^{-1}$. Therefore, the truncated eigenvector of Y is regarded as the class subspace Y_{c^t} at $\xi(m, D)$. Here, the Y_{c^t} with m orthonormal basis vectors represents the prominent scattering characteristics of the class c^t. Considering the unstable and inaccurate target cues represented by the mean of few-shot samples, we directly obtain Y_{c^t} by using the original semantic features from f_ϕ^t without subtracting their means for normalization. Thus, class-entity signals formulated as target structural cues at various azimuth angles can be represented by subspaces. Also, different classes are represented as separated subspaces in our method to make intraclass azimuth-aware features compact while avoiding inference among interclass ones.

3) Subspace Classification: The subspace classification, formulated as (2), is conducted by finding the closest distance

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between the original features $f_\phi(x_i)$ of a testing sample x_i and its projections at subspace S_c of class c modulated by the coefficient α. A differentiable closed-form solution to (2) can be written as (3) [48], where P_c is the transformation matrix of S_c and is used to project $f_\phi(x_i)$ onto the subspace of class c spanned by the column vectors of S_c. In our method, as shown in the bottom left of Fig. 3, the projection metric is employed for classification at $\xi(m, D)$. Here, $f_\phi(x_i)$ refers to target semantic features extracted by the backbone f_ϕ, and the transformation matrix P_c is computed as follows:

$$d\left(f_\phi(x_i), S_c\right) = \min_a d\left(f_\phi(x_i), S_c a\right) = \min_a \left\| f_\phi(x_i) - S_c a \right\|_2 \tag{2}$$

$$d\left(f_\phi(x_i), S_c\right) = \left\| f_\phi(x_i) - P_c f_\phi(x_i) \right\|_2 \tag{3}$$

$$P_c = Y_c Y_c^\top. \tag{4}$$

Then, probabilities of the sample x_i assigned with the class c can be calculated by applying a softmax function on all distances of the $f_\phi^t(x_i)$ to all class subspaces, where $\cup_{i=1}^t \mathcal{C}^i$ refers to all seen classes until session t

$$p(c|x_i) = \frac{\exp^{-d\left(f_\phi^t(x_i), S_c\right)}}{\sum_k^{\cup_{i=1}^t \mathcal{C}^i} \exp^{-d\left(f_\phi^t(x_i), S_k\right)}}. \tag{5}$$

In contrast to identifying targets by canonical point-to-point distances, such as L_2, cosine, or linear classifiers, we formulate the classification as a point-to-space matching problem by using subspace classifiers. Specifically, benefiting from the preservation of complete knowledge in class-aware subspaces, intrinsic patterns and relationships of targets at various imaging conditions can be fully captured, improving our method's ability to discriminate targets accurately and stably.

E. Losses Descriptions

1) Semantic Margin Separation: For the model's plasticity, distinct separations between new and old classes at semantic feature spaces should be maintained primarily. In particular, due to the scarcity of target-discerning information provided by few-shot samples, target characteristics of new classes extracted by a model trained by a CE loss are easily biased with a weak generalization, also causing a severe distortion on feature spaces of learned categories.

Considering a tight relationship between phase awareness of semantic features under the cosine criterion and target azimuth-aware structures, an SMS loss is proposed to learn new concepts adaptively based on specific sample pairs. As formulated in (6), with x^a selected from $\mathcal{D}^t$ as a compared anchor, x_i is a positive sample, of which the cosine distance, calculated as (7), to x^a is the farthest among the same class samples. x_j is a negative sample, of which the distance to the anchor is the nearest among all examples of other classes. $\hat{m}$ is a coefficient controlling the separations between positive and negative pairs. The larger the value of $\hat{m}$, the more distinct difference between old and new classes is preferred. The SMS loss can be diagramed in Fig. 5. The model trained by the SMS loss can learn new concepts while avoiding unnecessary parameter updates

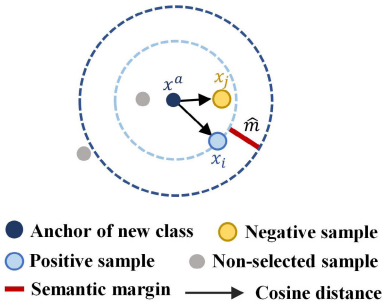


Fig. 5. Illustration of SMS loss.

$$\mathcal{L}_{\text{SMS}} = \sum_{x_j}^{E^{t-1}\cup\mathcal{D}^t} \sum_{x^a, x_i}^{\mathcal{D}^t} \max(0, \hat{m} - d(x^a, x_i) + d(x^a, x_j)) \tag{6}$$

$$d(x^a, x_i) = 1 - \langle \overline{f_\phi}(x^a), \overline{f_\phi}(x_i) \rangle, \overline{f_\phi}(x) = f_\phi(x) / \left\| f_\phi(x) \right\|_2. \tag{7}$$

2) Deep Subspace Separation: According to (1), the cosine-based principle angle $\cos\theta_k$ is measured by the similarity of two basis vectors from different subspaces, implicitly suggesting the ordered correlations of significant structural cues of different classes. The smaller the $\cos\theta_k$, the more distinct the difference between compared classes. Therefore, it is naturally preferable to improve the model's discriminability by separating different subspaces from the view of the principle angle. We leverage a DSS loss, in which the projection metric is considered based on the Grassmannian geometry. Originally, the distance d_P between two subspaces Y_j and Y_k at the Grassmannian manifold is defined as (8), which is the L_2 norm of the sine of principle angles. Specifically, the projection metric can be understood by assigning a point $\text{span}(Y) \in \xi(m, D)$ with its projection matrix $YY^\top \in \mathbb{R}^{D\times D}$, the distance can be formulated as an Euclidean distance defined in (9). F is the Frobenius norm. Notably, enlarging the d_P is equivalent to minimizing the second term of the formula

$$d_P\left(Y_j, Y_k\right) = \left(\sum_{i=1}^m \sin^2\theta_i \right)^{1/2} = \left(m - \sum_{i=1}^m \cos^2\theta_i \right)^{1/2} \tag{8}$$

$$d_P\left(Y_j, Y_k\right) = \left\| Y_j Y_k^\intercal - Y_k Y_j^\intercal \right\|_F^2 = 2m - 2\left\| Y_j Y_k^\intercal \right\|_F^2 \tag{9}$$

$$\mathcal{L}_{\text{DSS}} = -\sum_j^{\mathcal{C}^t} \sum_k^{\cup_{i=1}^{t-1}\mathcal{C}^i} d_P\left(Y_j, Y_k\right), \quad j \neq k. \tag{10}$$

In (10), the DSS loss is formulated as the sum of projection distances between subspaces of new classes $\mathcal{C}^t$ to all learned ones $\cup_{i=1}^{t-1}\mathcal{C}^i$ until the session t. As all base vectors of subspaces are compared within the distances, class-entity signals acted as class-aware separation supervisions can be considered to adaptively optimize the model.

3) Structure Less Forgetting: Due to the nonuniqueness of matrix decomposition and the limited target details represented by subspaces, overcoming catastrophic forgetting on the manifold is intractable and insufficient. In particular, as the target partial and global scattering patterns are mainly possessed

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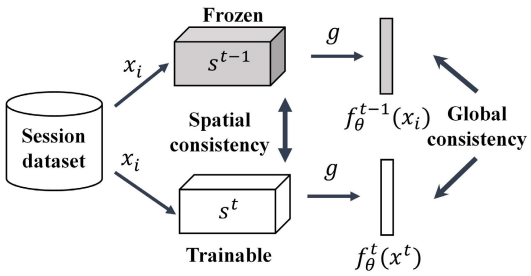


Fig. 6. Illustration of SLF loss.

by their spatial and aggregated features, an SLF loss with both spatial and global constraints is developed for knowledge preservation

$$\begin{aligned}\mathcal{L}_{\text{SLF}} &= \mathcal{L}_{\text{sp}} + \mathcal{L}_{\text{gap}} \\ &= \|s^{t-1}(x_i) - s^t(x_i)\|_F^2 + \left\langle f_\theta^{t-1}(x_i), f_\theta^t(x_i)\right\rangle \\ &\quad \text{where } x_i \in E^{t-1} \cup \mathcal{D}^t. \end{aligned} \tag{11}$$

As shown in Fig 6, semantic features $f_\phi^t(x_i)$ of a sample x_i can be decomposed as $f_\phi^t(x_i) = g(s^t(x_i))$. Here, $s^t(\cdot)$ composed of a stack of convolutions is used to extract target spatial scattering patterns. g is a global-average-pooling (GAP) operation for feature aggregation. In (11), the SLF loss is designed to punish both spatial and global drift of target features mapped by the frozen f_ϕ^{t-1} and the trainable f_ϕ^t models via a Frobenius and an inner product norms, respectively. Notably, as the point-to-point drifts and relative structures (angles) are constrained by the Frobenius and the inner product norms separately, both the model's representation ability to target local patterns and its flexibility to new ones can be maintained, balancing the stability and plasticity dilemma to some extent.

F. Azimuth-Aware Exemplar Selection

Since the knowledge-distillation paradigm our method follows, selecting appropriate exemplars could provide more precise supervision signals for knowledge preservation. Specifically, as the pronounced **volatility** and **periodicity** of intraclass targets in SAR images, two criteria, including **representativeness** and **complementarity**, should be obeyed for better knowledge preservation. Here, a class from the airborne MSTAR dataset is employed for clear demonstration, considering its completeness in azimuth angles.

1) *Representativeness:* Exemplars satisfying the criterion usually have high energy and contain widely target-discerning cues, which can lead to higher correlations to intraclass samples observed at diverse azimuth angles. Intuitively, both pixel-level (structural) and feature-level (semantic) similarity scores between five-compared anchors and their intraclass samples at full azimuth angles are presented as Fig. 7(a) and (b), respectively. Here, the two types of curves are with a similar fluctuating trend. However, as shown in Fig. 7(b), a high-energy anchor with salient and common backscattering cues can obtain a more competitive average score than a weak-energy one. For example, the score is 0.743 for the anchor at 270° azimuth angle but is 0.644 for that at 0° azimuth angle.
2) *Complementarity:* Exemplars satisfying the criterion could cover more infrequent target cues without involving much redundancy, providing complete information for knowledge preservation. In Fig. 7(b), the correlation scores ($Corr$) of the target at 0° azimuth angle to the other four anchors are computed and shown as black arrows. Here, even though the weaker energy of the target at 0° azimuth angle compared with that at 90° azimuth angle, the correlation curve of the target at 0° azimuth angle is approximately inverse compared with that of the latter, implicitly demonstrating the complementary structures provided by targets observed at apparently different postures.

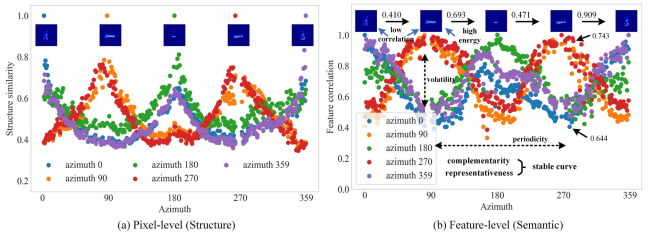


Fig. 7. Volatility and periodicity of intraclass targets at both pixel (structural) and feature (semantic) levels. The two types of similarity curves are obtained by comparing five target anchors with greatly different azimuth angles with intraclass ones at full postures. (a) Pixel-level (structure) similarity. (b) Feature-level (semantic) similarity. Both complementarity and representativeness criteria should be obeyed for exemplar selection.

Under the guidance of the above-mentioned properties, an AES strategy is designed, considering the links between the properties and target energy and correlations. Algorithm 2 details the strategy for selecting exemplars of a class c from $\mathcal{D}^t$ at the session t.

IV. Experiments

In this section, datasets for FSCIL of SAR ATR are first detailed. Next, implementation details, evaluation metrics, and compared benchmarks are introduced. Then, detailed investigations on the effects of subspace classifiers, the proposed losses, and the exemplar selection strategy are provided by ablation studies. Extensive results of our method compared with other methods are provided finally.

A. Dataset Description

In experiments, three FSCIL of SAR ATR datasets are constructed from two public and a self-collected datasets following Fig. 8, separately.

Overall, classes of the original dataset are randomly divided into two parts for base and incremental session learning. The related training and testing samples are also prepared in advance. For base session classes, abundant training samples covering wide azimuth angles (postures) are provided as $\mathcal{D}^1$. For incremental session classes, only K instances for each

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Algorithm 2 AES Strategy

Input: The trained f_ϕ^t, training data $X = \{x_1, x_2, .., x_n\}$ of class c, $X \in \mathcal{D}^t$, $c \in \mathcal{C}^t$

Require: m target number of exemplars

for $k = 1,\ldots,m$ **do**

Step 1: select the first (k=1) exemplar with the highest energy.

$e_1 \leftarrow \arg\max\{\| f_\theta^t(x_i^t)\|_2^2\}_{i=1}^n$

$E_c^t \leftarrow (e_1)$ //update exemplar set

Step 2: select other exemplars

Compute $\hat{\mu} \leftarrow (1/k - 1)\sum_x^{E_c^t} f_\theta^t(x)$ //current exemplar mean

$e_k \leftarrow \arg\max_{x \in X \setminus E_c^t} (\exp(-\frac{Corr(f_\theta^t(x), \hat{\mu})}{k}) \cdot \| f_\theta^t(x)\|_2^2)$

$E_c^t \leftarrow (e_1, e_2, \ldots, e_k)$//update exemplar set

end for

Output: Exemplar set E_c^t of class c

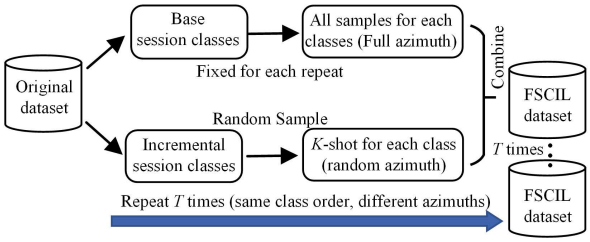


Fig. 8. Diagram of constructing the FSCIL dataset.

one are randomly selected from the original training samples. Finally, an FSCIL dataset can be created by combining samples of base and incremental classes together. In particular, the total of T individual FSCIL datasets are obtained by repeatedly sampling incremental class samples for stable evaluation. By splitting new classes into different groups, settings with various sessions can be formed. In default, only a new class is observed at each incremental session. The details of the datasets are provided as follows.

1) MSTAR-FSCIL Dataset: As an authorized airborne SAR ATR dataset, the MSTAR dataset contains various types of armored ground vehicles imaged at the X-band spotlight mode with 0.3-m resolution under various conditions. In experiments, targets of classes imaged at the standard operation condition (SOC) are used for construction. Specifically, targets at 17° and 15° depression angles are used for training and testing, respectively. For base session data $\mathcal{D}^1$, four classes, i.e., BTR70, 2S1, BRDM2, and BMP2, imaged at the 17° depression angle with azimuth angles covering 0° to 360° are selected. The remaining six classes, i.e., ZIL131, T62, D7, BTR60, T72, and ZSU234, under the same imaging conditions as the base are regarded as incremental session data $\mathcal{D}^t (t > 1)$. The targets of the dataset in optical and SAR images are shown in Fig. 9. The detailed configurations are given by Table I.

2) SAR-AIRcraft-1.0-FSCIL Dataset: SAR-AIRcraft-1.0 [49], a recently published spaceborne dataset for aircraft detection in SAR imagery, contains 4368 slices cropped from large-scale high-resolution SAR images imaged by the Chinese GF-3 satellite with a single polarization and spotlight mode.

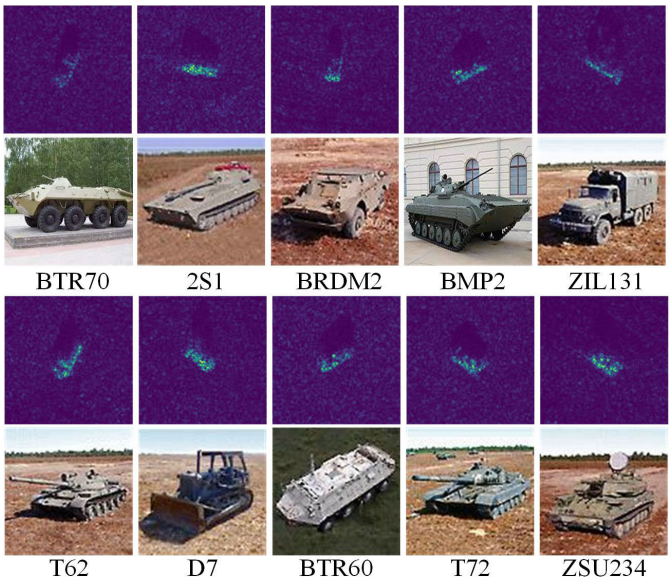


Fig. 9. Ten-class targets of the MSTAR dataset in SAR and optical images. (Best viewed in pseudocolor for target SAR images.)

TABLE I

CONFIGURATIONS OF THE MSTAR-FSCIL DATASET

Session	Order	Type	Serial No.	Train	Test
Base	1	BTR70	c71	233	196
	2	2S1	b01	299	274
	3	BRDM2	E-71	298	274
	4	BMP2	9563	233	196
Incremental	5	ZIL131	E12	5	274
	6	T62	A51	5	273
	7	D7	92v13015	5	274
	8	BTR60	k10yt7532	5	195
	9	T72	132	5	196
	10	ZSU234	d08	5	274

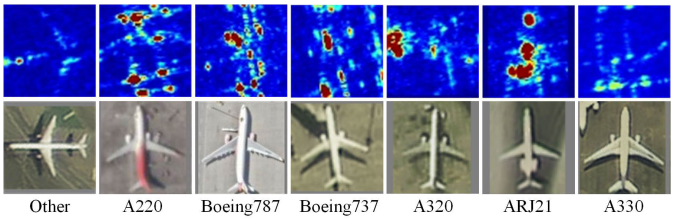


Fig. 10. Seven-class targets of the SAR-AIRcraft-1.0 dataset in SAR and optical images.

The spatial resolution of images is about 1 m. The dataset contains the number of 16 463 aircraft with categories covering A220, A320/321, A330, ARJ21, Boeing737, Boeing787, and others. To construct the FSCIL dataset, four types with the largest number of instances are served as base session classes. The remaining six are used for IL. For a fair comparison, we randomly select 2000 and five samples from the original dataset for base and incremental session training. The number of testing samples is 200 for all types of aircraft. Fig. 10 gives all types of SAR and optical images. Table II details the configurations of the constructed dataset.

3) Self-FSCIL Dataset: Apart from the FSCIL datasets built on the above-published dataset, another dataset derived from self-collected real spaceborne imagery is also constructed for further comparison. The original data contains 13 large scenes

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TABLE II
CONFIGURATIONS OF THE SAR-AIRCRAFT-1.0-FSCIL DATASET

Session	Order	Type	Train	Test
Base	1	Other	2000	200
	2	A220	2000	200
	3	Boeing787	2000	200
	4	Boeing737	2000	200
Incremental	5	A320	5	200
	6	ARJ21	5	200
	7	A330	5	200

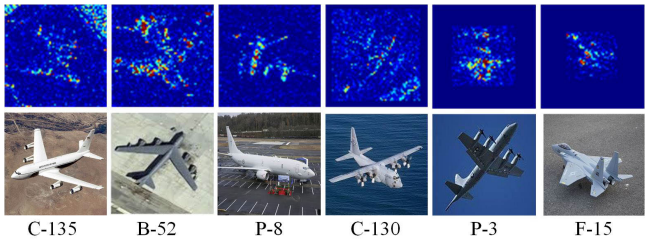


Fig. 11. Six-class targets of the self-collected dataset in SAR and optical images.

TABLE III
CONFIGURATIONS OF THE SELF-FSCIL DATASET

Session	Order	Type	Train	Test
Base	1-4	SAR-AIRcraft-1.0-FSCIL	2000	20
Incremental	5	C-135	5	20
	6	B-52	5	20
	7	P-8	5	20
	8	C-130	5	20
	9	P-3	5	20
	10	F-15	5	20

with five airports captured by the Chinese Chaohu-1 and Haisi-1 commercial earth observation satellites in spotlight mode with 1-m spatial resolution. The two missions with C-band SAR payloads are launched in 2020 and 2022 separately. After carefully annotated by SAR experts, six categories of aircraft, containing C-135, B-52, P-8, C-130, P-3, and F-15 with 100, 95, 49, 47, 31, and 31 instances, are obtained and merely used for IL because of the limited samples. Alternatively, the categories with abundant samples from the FSCIL dataset on the SAR-AIRcraft-1.0 are used for base session learning, considering the near imaging conditions. Besides, only 20 samples are randomly selected for evaluation. Fig. 11 shows all types of aircraft from self-collected images in SAR and optical formats. Table III details the configurations of the constructed dataset.

B. Implementation Details

Following topology-preserving knowledge incrementer (TOPIC) [39], the original paper of the FSCIL, we randomly select five-shot samples for each incremental class and repeat $T = 10$ times to obtain statistic performance. ResNet-18 [50] pretrained on the ImageNet [51] is acted as the feature extractor f_ϕ for all compared methods. Only two exemplars are selected by the AES. Hence, the m and D of $\xi(m, D)$ are 2 and 512, respectively. The class margin $\hat{m}$ of the SMS is 0.7. At the base session, all these methods are trained by the stochastic gradient descent (SGD) optimizer for 50 epochs with a mini-batch size of 32. The initial learning rate is 1e-2 and is decayed by 0.1 at epochs of 30 and 40. For IL, the model is trained for 100 epochs. The initial learning rate is 1e-4 and decayed by 0.1 at 45 and 70 epochs. The input images are normalized into 0 to 1 and cropped with sizes of 64 × 64 pixels, where targets are located centrally. We randomly rotate the targets with small degrees, e.g., 10° in this article, for data augmentation based on the SAR imaging mechanisms. All the experiments are conducted on a server equipped with NVIDIA GTX 3090 graphic cards with CUDA 11.1 support.

C. Evaluation Metrics

Three metrics are employed to judge the performance of compared methods. The classification accuracy (Acc.) evaluates the performance of compared methods for classifying all encountered classes at each learning session. The averaged accuracy (Avg.Acc) comprehensively evaluates the model's performance by averaging the Acc results of all sessions. The larger the Avg.Acc, the better the model could learn continually from few-shot samples. The performance (PD) drop rate [43] measures the absolute PD of the last trained models, which is calculated by subtracting the accuracy of the last session from that of the first session. The smaller the PD, the better the model could overcome the catastrophic forgetting issue. There is no intersection among models learned on different sequences. In a specific sequence, the model learned at the former sessions is directly used for the subsequent ones. The performance of a comparative method is obtained by statisticizing its session results on all the sequences.

D. Benchmark Descriptions

Numerous task-specific approaches are employed for evaluation, which are described as follows.

1) *Traditional CNN-Based Methods:* Fine-tuned CNN (Ft-CNN)—directly fine-tunes a CNN-based classifier on the current session dataset using a CE loss. Oracle—jointly trains all classes offline by a CE loss, which can be viewed as the upper bound of compared methods.
2) *IL-Based Methods:* iCaRL [52]—utilizes an external nearest class mean as the final classifier and preserves the learned knowledge by using knowledge distillation and herding techniques. End-to-end incremental learning (EEIL) [53]—constructs an end-to-end learning manner and improves the iCaRL by introducing a cross-distilled loss function and a balanced fine-tuning technique. LUCIR [47]—learns a unified classifier to tackle the instance-imbalanced problem at the cosine space. These approaches consider the catastrophic forgetting issue by saving old class exemplars for knowledge distillation during IL.
3) *FSL-Based Methods:* ProtoNet [29]—learns a metric space by episodic training and performs classification by computing distances of query samples to the prototypical representation of each class. MatchNet [54]—is designed for one-shot learning in which a bidirectional long short-term memory (BiLSTM) [55] is used to encode contextual information of the support and the query samples for metric learning.

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TABLE IV
QUANTITATIVE RESULTS OF DIFFERENT CRITERIA FOR FSCIL OF SAR ATR (ONE-WAY FIVE-SHOT SETTING)

Train/Test criteria	Acc. in each session (%)							Avg.Acc (%)	PD (%)
	1	2	3	4	5	6	7		
$L_2 \rightarrow L_2$	94.61	79.41	66.24	61.87	55.91	50.48	45.06	64.80	49.55
$L_2 \rightarrow$ Subspace	94.60	81.07	70.79	66.42	60.03	54.30	49.37	68.08	45.23
FC$\rightarrow L_2$	88.71	76.80	66.55	58.90	57.84	53.10	47.56	64.21	41.15
FC$\rightarrow$ Subspace	99.65	85.17	70.11	64.49	61.11	57.26	52.24	70.00	47.41
Cos$\rightarrow$ Cos	99.70	84.48	69.17	63.89	59.94	56.31	51.24	69.25	48.46
Cos$\rightarrow$ Subspace	98.59	85.72	74.10	65.12	62.04	57.77	51.66	70.71	46.93

RelationNet [30]—learns a relation module by the episodic training and classifies targets by computing relation scores along feature channels between query images and class prototypes supported by limited examples. These methods mainly learn metric spaces using the meta-learning technique and classify targets based on well-designed matching strategies.

4) *FSCIL-Based Methods:* TOPIC [39]—first solves the problem by preserving learned knowledge through a neural gas network [56]. Exemplar relation distillation incremental learning (ERDIL) [42]—proposes an exemplar relation graph and distills relational features to prevent forgetting. Incremental deep learning vector quantization classification (IDLVQC) [57]—compresses class knowledge into a small number of quantized reference vectors and learns continually by adding more vectors sequentially. CEC [43]—introduces a continually evolved classifier in which a graph attention network is trained episodically through a pseudo-IL stage to adjust class embeddings and features of testing samples adaptively. Forward compatible training (FACT) [58] achieves SOTA performance by designing a forward-compatible model in which virtual prototypes are assigned to squeeze the embedding of known classes and reserve for new ones.

For fair comparisons, the comparative methods are trained by the same general settings, e.g., mini-batch size, learning rate, and model initialization, as those for our approach. Other hyperparameters are used following the original papers. Especially, for the FSCIL methods, the number of NG nodes in TOPIC is the same as that of the saved exemplars for old and new classes. In CEC, only a category with a single sample is episodically selected from the base session dataset to formulate a one-way one-shot setting at the pseudo-IL stage, considering the limited number of base classes. In FACT, six virtual prototypes are preassigned in the embedding space following the strategy of the original paper.

E. Ablation Study

In this section, the contributions of the subspace classifier and the proposed strategies and losses are verified successively. In particular, the derived MSTAR-FSCIL dataset is leveraged for investigation, considering its advantages of target azimuth completeness and category diversity.

1) Effects of Subspaces: To purely verify the effectiveness of subspaces for FSCIL of SAR ATR, the network f_ϕ is merely trained on the $\mathcal{D}^1$ by three typical criteria, including Euclid, fully connection (FC), and cosine (Cos), and is frozen to extract target patterns at later sessions. We compare the session performance produced at the original semantic space and the manifold. The former results are obtained by the nearest class-mean (NCM) classifier by comparing target features with class prototypes. The latter can be obtained following Section III-D3.

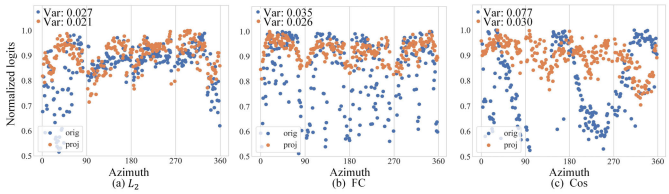


Fig. 12. Intraclass compact predictions obtained by subspace classifiers at all target azimuth angles. Predictions obtained by the three metrics at original spaces are notated as the orig, and those by subspace classifiers at the projected spaces are notated as the proj. Logits representing original responses are normalized for clear comparisons. Intraclass stability is quantified as the variance of logits at the top left of subfigures. (a) L_2. (b) FC. (c) Cos.

1) *Overall Judgment:* Table IV reports the quantitative results. The training and testing metrics are shown on the left and the right sides of the $\rightarrow$, respectively. The incremental-frozen f_ϕ is tested by the classical criteria and the subspace classifier alternately. From the first, third, and fifth columns of Table IV, benefiting from the special consideration of phase information for representing target azimuth-aware cues, the f_ϕ^1 learned by the cosine criterion obtains the most competitive performance compared with those by the L_2 and the FC. Furthermore, from the second, fourth, and last columns of Table IV, the method's performance on most sessions can be further improved by using the subspace classifier, especially for the model learned by the cosine criterion, roughly demonstrating its superiority for target classification by leveraging few-shot samples.

2) *Intraclass Compactness:* The compactness of intraclass target features at the manifold is verified in this part. In Fig. 12, the ground-truth (GT) logits of testing samples predicted at the original (orig) and the projected (proj) spaces are plotted in blue and orange scatters. The variance values for the two types of scatters are given in the top-left figures. Logits at the two spaces are normalized for fair comparisons. It can be seen that the logit scatters at the projected space (the orange scatters) are more compact with lower variance than those at the original space (the blue scatters) as target azimuth changes, verifying the stability and compactness of subspaces for representing features of intraclass targets at diversified postures.

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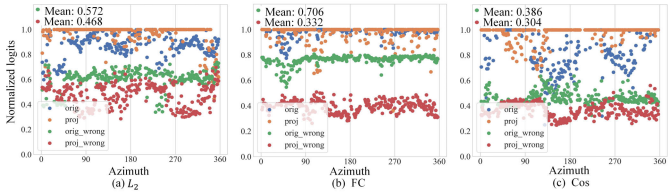


Fig. 13. Interclass discrimination obtained by subspace classifiers at all target azimuth angles. The GT logits obtained by the three metrics at original spaces and by the subspace classifier at the projected spaces are notated as the orig and the proj separately. The average non-GT logits obtained at the orig and the proj are plotted are notated as the orig_wrong and the proj_wrong, separately. Interclass separation is quantified as the average of GT and non-GT logits at the top left of subfigures. (a) L_2. (b) FC. (c) Cos.

3) *Interclass Separation:* For interclass separation, GT and non-GT logits predicted at the original (scatters in blue and green), and the projected spaces (scatters in orange and red) are shown in Fig. 13. Logits of non-GT labels are averaged for overall comparison. The mean of GT and non-GT logits of targets at all azimuth angles are provided in the top-left figures. It can be seen that the GT and the non-GT logits at the projected spaces maintain more distinct separations with a lower mean of non-GT logits than those at the original, suggesting the advantages of the subspace classifier for extracting class-aware features and discriminating interclass ones accurately. Furthermore, the average distances of class bases between new and old classes and those among all categories are shown in Fig. 14(a) and (b), respectively. It can be seen that both two types of distances calculated by class subspaces are farther than those calculated by the original ones, suggesting clear separations among different class subspaces.

4) *Background Robustness:* With the features produced by the intransigent model f_ϕ^1 under the cosine criterion as an example, the background robustness of projected features is verified and shown in Fig. 15. With the degradation of image quality, some conclusions can be provided as follows. In Fig. 15(a), it can be seen that the model's performance on old, new, and all classes can be stably improved by classification at the projected spaces. Furthermore, the class separability quantified as the L_2 distance of features and the variance of intraclass features are shown in Fig. 15(b) and (c), respectively. It can be seen that the farther distances among interclass features and the lower variance of intraclass ones can be maintained by classification at the projected spaces, even with the degeneration of image quality, verifying the stability and robustness of subspace classification.

2) Effects of AES: Since our method follows the exemplar-based paradigm, selecting representative exemplars is crucial for knowledge preservation and discrimination at later sessions. Overall, as shown in the second and third rows of Table V, compared with the method without the AES, our method with the AES achieves obvious performance gain on the Acc. at most learning sessions and the Avg.Acc. Fig. 16 shows the incremental classification performance of our method on base and all seen classes with the support of different numbers of exemplars selected by the random,

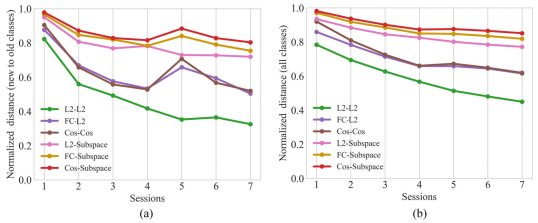


Fig. 14. Average distances among class bases at the original and the projected spaces at each session. Distances are normalized based on ranges at their metric spaces. (a) Average distances of the bases of new classes to the old. (b) Average distances of the bases of all classes.

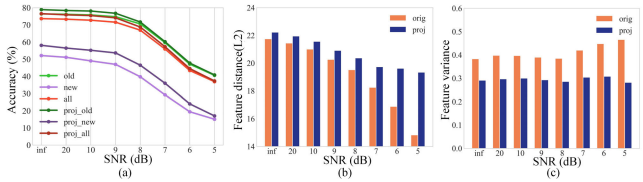


Fig. 15. Robustness of classification by the cosine and the subspace classifiers at the original (orig) and the projected (proj) spaces under different SNR levels. (a) Classification accuracy of the old, new, and all classes averaged from all learning sessions. (b) Average distances of different class features at the original and the project spaces under different SNR levels. (c) Variance of the same class features at the original and the project spaces at different SNR levels.

the herding [52], and the AES strategies. Notably, exemplars chosen by the herding are always near the class centers. Some insights can be provided as follows.

Overall, as shown in Fig. 16(a), with any of the strategies, the classification accuracy of our method on all classes shows a stable improvement with the increase of saved exemplars. In particular, our method with the AES achieves the highest performance on both base and all classes at all conditions of exemplar numbers, and it achieves a higher performance gain on base classes than those with the other two strategies. Thanks to more suitable exemplars selected by the AES than those selected by random and herding, more representative and discernible information can be preserved, providing more stable and accurate decision boundaries for target identification. Furthermore, the predicted logits and the corresponding variance are provided in Fig. 16(b) and (c), respectively. Since class subspaces are constructed by saved exemplars, the higher logits with lower variance indicate the more accurate matching capability and stability between class-aware subspaces and testing samples. Holistically, as the number of exemplars increases, the responses of GT logits increase with a stable decrease in the variance. Thanks to both general and particular azimuth-aware structures covered by exemplars selected by the AES, the highest responses and the lowest variance on both base and overall classes are achieved. Although exemplars selected by the herd strategy are centralized, they are suboptimal in representing diversified target structures due to the low diversity. Furthermore, feature correlation curves of class exemplars are plotted in Fig. 16(d). It can be seen that exemplars selected by the AES strategy have the lowest similarity scores than those by other strategies, verifying complementary target features considered by the AES for knowledge retention.

3) Effects of SMS: As shown in the fourth row of Table V, a remarkable improvement in classification accuracy of all

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TABLE V
CONTRIBUTIONS OF COMPONENTS OF OUR METHOD TESTED ON MSTAR DATASET UNDER ONE-WAY FIVE-SHOT SETTING

Stability		Plasticity		Subspace	Acc. in each session (%)							Avg. Acc	PD ↓
AES	SLF	SMS	DSS		1	2	3	4	5	6	7		
					99.70	84.48	69.17	63.89	59.94	56.31	51.24	69.25	48.46
				✓	98.59	85.72	74.10	65.12	62.04	57.77	51.66	70.71	46.93
✓				✓	99.89	86.75	73.01	67.72	64.59	59.89	54.79	72.38	45.10
✓		✓		✓	99.89	89.15	75.09	71.63	69.42	63.84	57.87	75.27	42.02
✓	✓	✓		✓	99.89	88.89	76.23	72.36	71.36	65.8	60.43	76.42	39.46
✓	✓	✓	✓	✓	99.89	89.73	76.33	73.64	71.46	66.97	62.31	77.19	37.58

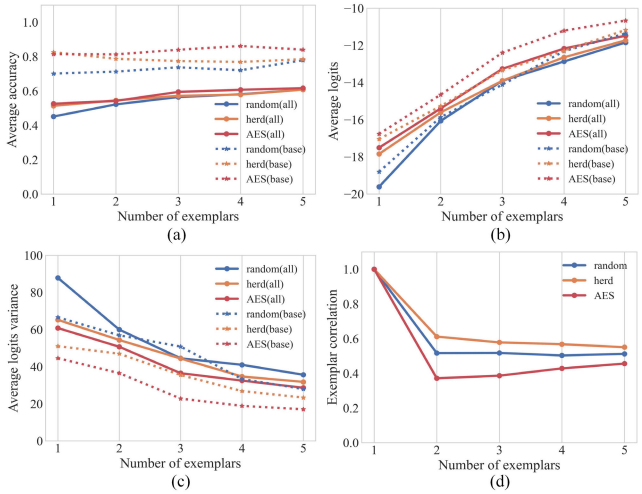


Fig. 16. Superiority of the AES compared with two classical exemplar selection strategies at various exemplar numbers. (a) Average accuracy of the base and all seen classes. (b) Average logits of the base and all seen classes. (c) Average logit variance of the base and all seen classes. (d) Correlations of class exemplars selected by different strategies.

learning sessions is achieved by our method with the SMS. The Avg.Acc and the PD of our method reach 75.27% and 42.02%, which are 2.89% higher and 3.08% lower than the method without the SMS. Furthermore, the performance of our method learned by different loss variants is shown in Table VI. Apart from the Avg.Acc and the PD, the average accuracy on old, new, and all classes of all incremental sessions is reported and marked as Inc.Acc (old/new/all). Since no classifier is adopted during IL, a proxy linear classifier is attached above the f_ϕ for the 'AASC w/ CE'. The compared anchors in the "AASC w/ SMS(all)" are from old-class exemplars and current session data. Accordingly, some conclusions can be summarized as follows. First, the "AASC w/ SMS" achieves the highest performance at all indicators among all variants. Second, the performance gains on new classes (Inc.Acc new) obtained by the "AASC w/ SMS" and the "AASC w/ SMS(all)" are more obvious than that by the "AASC w/ CE." Due to the overlook of special separations among interclass samples with similar scattering patterns, class boundaries guided by the CE loss are blurry. However, benefiting from enlarging pairwise discrepancies between new and old classes, discerning and detailed features of new classes can be specially considered for judgment by the SMS and its variants. Third, a higher performance gain on old classes is achieved by the "AASC w/ SMS" than by the "AASC w/ SMS(all)." Although margins between pairwise samples of different classes are constrained by the SMS(all), old-class decision boundaries adjusted by the "AASC w/ SMS(all)" may

TABLE VI
MODEL'S PLASTICITY PROVIDED BY DIFFERENT SEPARATION LOSSES TESTED ON MSTAR DATASET UNDER ONE-WAY FIVE-SHOT SETTING

Method	Inc.Acc (%) ($t > 1$)			Avg.Acc (%)	PD (%) ↓
	old	new	all		
AASC w/o SMS	72.17	45.98	67.80	72.38	45.10
AASC w/ CE	72.90	51.03	69.18	73.56	42.33
AASC w/ SMS(all)	73.03	58.36	70.45	74.66	42.17
AASC w/ SMS	73.63	58.75	71.14	75.27	42.02

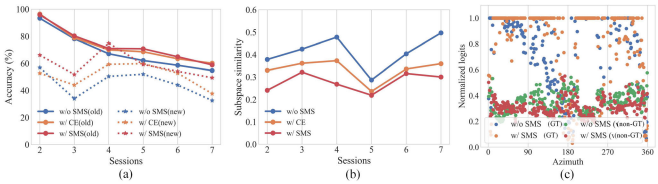


Fig. 17. Effects of the SMS loss on different learning sessions and target azimuth angles. (a) Classification accuracy on old and new classes. (b) Similarity of interclass subspaces. (c) Logits of the GT and non-GT labels predicted by subspace classifiers.

be easily distorted due to an excessive parameter's update from exemplars and compared anchors of old classes.

Intuitively, the effects of the SMS loss on different learning sessions and target azimuth angles are provided in Fig. 17. First, our method with the SMS loss achieves higher accuracy on novel classes without losing much performance on the old at each session. In Fig. 17(a), the dotted curve in red color is higher than those in blue and orange at most incremental sessions. Second, clearer separations of interclass subspaces are obtained by our method with the SMS loss than those with the CE or without the SMS. In Fig. 17(b), the lower similarity among interclass subspaces is preserved by our method with the SMS than other comparisons, even though serious fluctuations appear on the scores caused by unstable target cues provided by few-shot instances. Third, more discerning intraclass characteristics provided by few-shot samples can be fully represented by our method with the SMS loss, implicitly improving discrimination of class subspaces. The logits of testing samples on GT and non-GT labels, which are predicted by subspace classifiers, are provided in Fig. 17(c). In comparison with the GT and the non-GT logits (the blue and the green scatters) of features extracted by our method with the CE, the GT logits of features extracted by the model with the SMS (the orange scatters) are more stable with clearer separations to the non-GT ones (the red scatters) at all azimuth angles.

4) Effects of SLF: The contributions of the SLF loss are verified in this part. In the fifth row of Table V, distinct improvements on all metrics are achieved by our method with the SLF loss. For example, the Avg.Acc and the PD reach 76.42% and 39.46%, which are 1.15% and 2.56% superior

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TABLE VII
EFFECTS OF DIFFERENT POOLING DISTILLATION UNDER ONE-WAY FIVE-SHOT SETTING

Method	Inc.Acc (%) ($t > 1$)			Avg.Acc (%)	PD (%) ↓
	old	new	all		
SLF w/o $\mathcal{L}_{gap}$	74.35	61.86	69.00	76.08	39.64
SLF w/o $\mathcal{L}_{sp}$	74.80	59.18	72.25	76.19	40.10
SLF	74.80	61.63	72.51	76.42	39.46

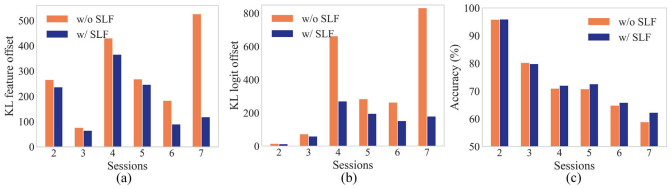


Fig. 18. Superiority of the SLF loss at three levels at each learning session. (a) Alleviation of feature drifts. (b) Alleviation of logit drifts. (c) Improvement in classification performance on old classes.

TABLE VIII
MODEL'S PERFORMANCE ON DIFFERENT TYPES OF CLASSES BY USING THE DSS LOSS

Method	Inc.Acc (%) ($t > 1$)			Avg.Acc (%)	PD (%) ↓
	old	new	all		
w/o DSS	74.80	61.63	72.51	76.42	39.46
w/ DSS	75.11	65.56	73.40	77.19	37.58

to the method before applying the loss (see the fourth row of Table V for an example), separately. Furthermore, the effects of loss components, i.e., $\mathcal{L}_{\text{sp}}$ and $\mathcal{L}_{\text{gap}}$, are reported in Table VII. Accordingly, compared with the method with the "SLF," a limited performance gain on the Avg.Acc is obtained by the method with the "SLF w/o $\mathcal{L}_{\text{gap}}$" or with "SLF w/o $\mathcal{L}_{\text{sp}}$." Specifically, due to a loose constraint on global patterns of old classes, the plasticity of the model with "SLF w/o $\mathcal{L}_{\text{gap}}$" for new concepts is more permissive, leading to a slight forgetting of old knowledge. However, due to a weak constraint on target local patterns provided by the method with "SLF w/o $\mathcal{L}_{\text{sp}}$," spatial semantic features between new and learned classes are easily ambiguous, implicitly hindering the model's ability to fast adapt to novel knowledge.

As the model evolves, old-class features and the predicted logits projected by models of different sessions are also changed. Thus, it is reasonable to measure the model's forgetting issue as drifts on target features and predicted logits of learned classes during IL. Specifically, the drifts are quantified as the Kullback–Leibler (KL) divergence on the two levels. The lower the KL divergence, the more old knowledge our method could preserve. As shown in Fig. 18(a) and (b), our approach with the SLF loss obtains a lower KL divergence than that without the loss on both feature and logit levels at all incremental sessions. In Fig. 18(c), even though the forgetting issue is gradually serious during IL, our method with the SLF loss achieves a higher performance in old classes, implicitly suggesting the advantages of the loss for knowledge preservation.

5) Effects of DSS: The DSS is designed for subspace orthogonality based on the Grassmannian geometry in our method. According to the numerical results shown in the last row of Table V, our method's performance is improved by

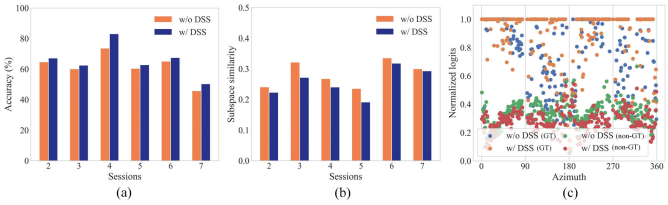


Fig. 19. Superiority of the DSS loss on incremental sessions and the intraclass predictions. (a) Improvement in the accuracy of new classes at incremental sessions. (b) Subspace similarity of the new to the old classes at incremental sessions. (c) GT and non-GT logits at all azimuth angles.

the DSS, and the superiority becomes much more pronounced as the number of sessions increases. Table VIII shows the performance on old, new, and all classes obtained by our method without or with the DSS. It can be seen that the results of our method with the DSS loss on these divisions are 0.34%, 0.75%, and 0.43%, higher than those without the loss, separately.

Apart from this, the effects of the DSS on different sessions and target azimuth angles are further investigated. In Fig. 19(a), serious fluctuations appear in the performance of novel classes due to diversified and unstable target cues provided by few-shot samples. However, the classification accuracy of new classes at all sessions is still promoted by the DSS loss. Besides, as shown in Fig. 19(b), subspaces formulated by our method with the DSS have lower similarity scores between old and new classes than those by our method without the DSS, suggesting clearer class separations by regularizing subspace orthogonality. Apart from the above-mentioned comparisons, logits of the GT and the non-GT labels for testing samples at all azimuth angles predicted by subspace classifiers are reported in Fig. 19(c). It can be seen that logits of the GT labels (the orange scatters) predicted by our method with the DSS are more stable with clearer separations to the non-GT ones than those by our method without the DSS. Thanks to special attention to the separability of bases between new-class subspaces and the old ones based on the Grassmannian geometry, more target-discerning cues can be captured for subspace construction, leading to more accurate projections of testing samples by subspace classifiers.

F. Benchmark Comparison

In this section, extensive experiments on the three FSCIL of SAR ATR datasets are conducted to comprehensively evaluate the performance of our method compared with various task-specific CNN-based benchmarks. The results and detailed analyses are provided in quantitative and qualitative facets.

1) Quantitative Analyses: The quantitative results of compared methods tested on three derived datasets with one-way five-shot setting, i.e., only a new category can be observed at each incremental session, are shown in Tables IX–XI, separately. In summary, our method has threefold advantages. First, our method achieves the topmost Avg.Acc compared with other solutions except for the oracle (the upper bound) on the three FSCIL datasets. Second, the catastrophic forgetting problem is well-addressed by our method, leading to a more competitive PD score than most approaches. Third, the stability and plasticity dilemma is well-balanced by our method at

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TABLE IX
COMPARISON WITH SOTAS ON MSTAR-FSCIL DATASET

Methods	Sessions (%)							Avg. Acc	PD ↓
	1	2	3	4	5	6	7		
Ft-CNN	98.94	76.21	62.01	52.02	47.55	43.08	38.12	59.70	60.82
Oracle	98.94	85.05	77.17	78.30	73.86	70.34	67.10	78.68	31.84
iCaRL [52]	92.12	80.37	69.35	61.16	58.02	53.47	47.75	66.03	44.37
EEIL [53]	98.94	87.46	73.14	64.99	63.56	57.98	52.25	71.19	46.69
LUCIR [47]	99.79	81.93	69.21	60.03	55.43	51.73	46.33	66.35	53.46
ProtoNet [29]	90.87	81.51	68.10	64.39	58.36	54.11	47.80	66.45	43.07
MatchNet [54]	70.34	57.57	47.03	46.69	42.01	37.89	32.87	47.77	37.47
RelationNet [30]	80.56	63.85	51.83	47.15	42.37	38.43	34.03	51.17	46.53
TOPIC [39]	91.10	85.27	72.77	63.25	61.24	56.03	50.03	68.53	41.07
ERDIL [42]	98.94	83.34	73.48	64.10	64.19	58.44	49.40	70.27	49.54
IDLVQC [57]	97.02	82.91	70.94	61.94	58.70	53.85	47.65	67.57	49.37
CEC [43]	90.54	80.52	72.27	72.53	66.97	61.76	57.97	71.79	32.57
FACT [58]	98.85	87.36	67.14	64.24	49.50	46.69	45.01	65.54	53.84
Ours (AASC)	99.89	89.73	76.33	73.64	71.46	66.97	62.31	77.19	37.58

TABLE X
COMPARISON WITH SOTAS ON SELF-FSCIL DATASET

Methods	Sessions (%)							Avg. Acc	PD ↓
	1	2	3	4	5	6	7		
Ft-CNN	99.99	79.70	63.67	54.71	46.44	43.56	39.75	61.12	60.25
Oracle	99.99	86.50	85.00	80.71	76.75	75.50	70.05	82.07	29.95
iCaRL [52]	98.75	94.60	80.25	68.14	55.56	52.39	48.05	71.11	50.70
EEIL [53]	99.99	96.40	85.75	74.79	65.94	63.50	58.85	77.89	41.15
LUCIR [47]	99.99	93.40	82.67	75.00	68.00	64.39	61.35	77.83	38.65
ProtoNet [29]	87.88	87.50	80.92	73.21	66.31	63.56	62.60	74.57	25.28
MatchNet [54]	28.50	31.20	28.92	24.57	21.38	23.17	22.55	25.76	5.95
RelationNet [30]	33.00	37.70	33.83	29.50	26.63	24.72	23.20	29.80	9.80
TOPIC [39]	99.38	92.90	81.50	72.36	64.13	59.67	56.15	75.16	43.23
ERDIL [42]	99.99	91.70	81.67	71.93	66.81	63.06	60.75	76.56	39.25
IDLVQC [57]	98.75	88.30	74.67	66.07	60.31	54.94	50.00	70.43	48.75
CEC [43]	74.88	73.90	67.08	62.71	57.06	57.17	56.20	64.14	18.68
FACT [58]	99.88	91.20	84.42	77.07	70.06	68.17	65.55	79.48	34.33
Ours (AASC)	99.99	96.00	86.17	77.71	70.00	68.28	64.30	80.35	35.70

TABLE XI
COMPARISON WITH SOTAS ON SAR-AIRCRAFT-1.0-FSCIL DATASET

Methods	Sessions (%)				Avg. Acc	PD ↓
	1	2	3	4		
Ft-CNN	99.62	82.93	67.66	59.32	77.38	40.30
Oracle	99.47	81.24	69.83	64.31	78.71	35.16
iCaRL [52]	99.50	88.07	74.42	63.21	81.30	36.29
EEIL [53]	99.62	87.56	73.54	65.67	81.60	33.95
LUCIR [47]	99.75	88.34	75.75	67.59	82.86	32.16
ProtoNet [29]	88.05	72.23	63.24	55.92	69.86	32.13
MatchNet [54]	30.01	24.90	20.57	18.41	23.47	11.60
RelationNet [30]	32.15	24.96	22.34	20.39	24.96	11.76
TOPIC [39]	99.64	89.09	76.95	67.06	83.18	32.58
ERDIL [42]	99.62	82.89	69.19	61.92	78.40	37.70
IDLVQC [57]	99.37	84.13	71.38	62.96	79.46	36.41
CEC [43]	78.97	65.64	57.76	51.75	63.53	27.22
FACT [58]	99.44	83.05	72.25	65.14	79.97	34.30
Ours (AASC)	99.50	89.12	77.18	69.28	83.77	30.22

intermediate sessions. The detailed analyses are provided as follows.

1) *Compared With Traditional CNN-Based Methods:* We report the performance of the conventional CNN-based classifiers in the first two rows of Tables IX–XI. Here, our method significantly outperforms the Ft-CNN and reaches the nearest even higher performance compared with the oracle. Due to the weak supervision from learned knowledge, a substantial forgetting problem occurs in the Ft-CNN. Although the competitive results of the Avg.Acc achieved by the oracle, i.e., 78.68%, 78.71%, and 82.07%, on three FSCIL datasets, the results (Acc.) at some special sessions (see the second session in Tables IX–XI for an example) are still unsatisfied, raised by its weak ability to learn base and new classes from target samples with a long-tail distribution.
2) *Compared With IL Methods:* The performance of the IL methods is shown from the third to fifth rows of Tables IX–XI. Here, the most competitive performance on three FSCIL datasets can be achieved by these methods, benefiting from numerous techniques designed for old-knowledge preservation and stable evaluation, e.g., exemplar-selecting mechanisms, knowledge distillation losses, balance fine-tuning (EEIL), and NCM classifier (iCaRL). Nevertheless, due to the difference in learning assumptions between the IL and the FSCIL, the capabilities of these IL methods for learning new knowledge from few-shot samples are still unlocked, easily exacerbating a biased learning of discerning features of base and incremental classes and leading to unsatisfied performance on intermediate sessions.
3) *Compared With FSL Methods:* The results of few-shot-learning (FSL)-based methods on the three datasets are reported from the sixth to the eighth rows of Tables IX–XI. Overall, a severe performance gap exists between the FSL methods and others, raised by the absence of powerful incremental designs. The reasons may be twofold. The first can be attributed to their task-agnostic property, which leads to limited utilization of new-class knowledge for updating interclass

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decision boundaries. The second comes from the lack of incremental techniques, making them fail to bridge the interclass connections at different sessions. In contrast, our method is task-specific and could maintain a global discrimination ability on both novel and old classes at all sessions, providing considerable capabilities of knowledge prompt transferring and stable preservation.

4) *Compared With FSCIL Methods:* The results of FSCIL approaches are shown from the ninth to the end of Tables IX–XI. The insights are twofold. First, most of them can achieve competitive performance on the three FSCIL datasets, thanks to their specially designed losses and strategies for alleviating the special stability and plasticity dilemma. Second, our method possesses a powerful adaptability to various FSCIL scenarios. Due to the difference in observing conditions, target configurations, and techniques, algorithms with the topmost performance are not consistent across the three FSCIL datasets, e.g., 71.79% for CEC in Table IX, 83.18% for TOPIC in Table XI, and 79.48% for FACT in Table X, which implicitly suggests their limited adaptability to wide scenarios of the practical task. However, our method with the subspace classifier and domain-knowledge-powered losses and strategies can fully capture azimuth-aware target-discerning cues for accurate inference, which leads to the best performance on all sessions compared with all benchmarks, even the oracle. For example, the scores of the Avg.Acc reach 77.19%, 83.77%, and 80.35% on these datasets captured by airborne and spaceborne platforms, respectively.

2) Qualitative Analyses: The qualitative performance of compared benchmarks is provided in this part. Considering various categories (vehicles, aircraft), imaging conditions (laboratory, real world), and platforms (airborne and spaceborne) covered by derived FSCIL datasets. The visual performance of methods on these datasets is reported for comprehensive judgment.

1) *Session Performance Curves:* The session performance of compared methods tested on the two derived MSTAR-FSCIL and the Self-FSCIL datasets is reported in Fig. 20. For each dataset, three scenarios with different settings (sessions), including one-way five-shot, two-way five-shot, and one-step five-shot, are formulated following Section IV. As the session evolves, some insights can be concluded as follows. First, caused by the catastrophic forgetting of old-class knowledge and inaccurate representation of the novel from few-shot samples, the performance of all compared methods encounters a constant decrease in all settings of each dataset. Second, benefiting from the special azimuth-aware losses, strategies and subspace classifier, our method (the red curve) stably obtains superior session performance in all learning scenarios (different datasets and settings) compared with all benchmarks even the oracle [see the red curves at the first three sessions in Fig. 20(a) and (b) for example], implicitly suggesting our method's strong adaptability to various FSCIL of SAR ATR situations. Third, methods for the same tasks

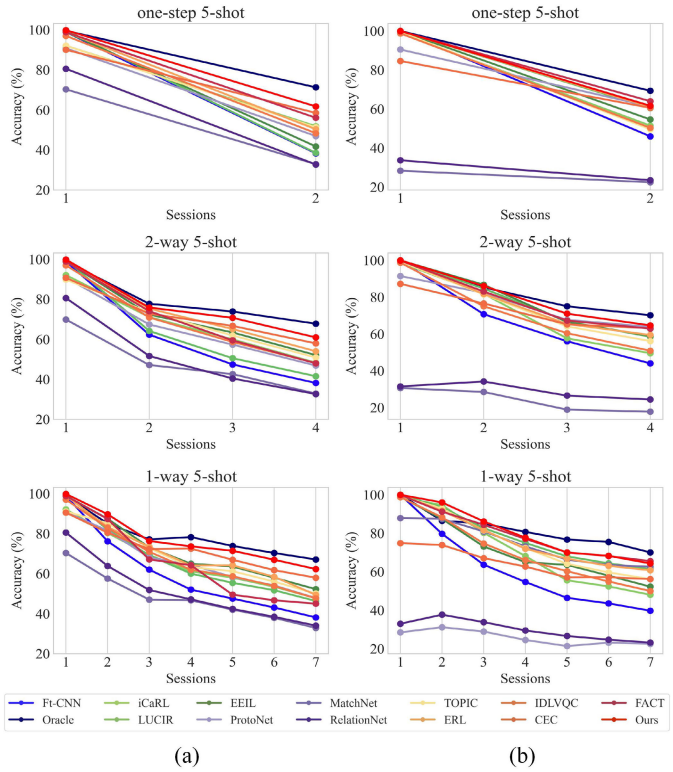


Fig. 20. Performance curves of compared benchmarks tested on two derived FSCIL datasets under three learning scenarios. (a) Performance curves on the MSTAR-FSCIL datasets. (b) Performance curves on the Self-FSCIL datasets.

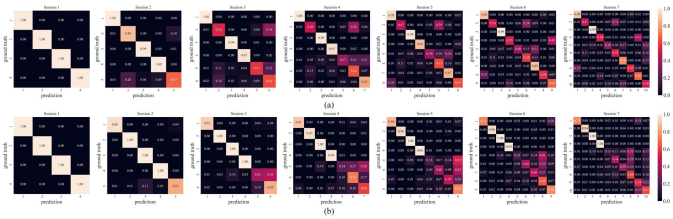


Fig. 21. Session confusion matrices of our method tested on the MSTAR-FSCIL and Self-FSCIL datasets with one-way five-shot setting. (a) Session confusion matrices of our method tested on the MSTAR-FSCIL dataset. (b) Session confusion matrices of our method tested on the Self-FSCIL dataset.

can usually achieve relatively similar performance in all scenarios. However, serious overlaps appear among those for different tasks. In particular, the performance of the Ft-CNN and the FSL-based methods is seriously unsatisfactory due to the lack of special techniques for continually learning new knowledge.

2) *Confusion Matrices:* The confusion matrices of our method and compared benchmarks tested on FSCIL datasets with one-way five-shot setting, derived from the MSTAR and the self-collections, are investigated in this part.

a) *Session Matrices:* Since different imaging conditions and targets within the MSTAR-FSCIL and the Self-FSCIL datasets, the sessionwise confusion matrices of our method tested on them are provided in Fig. 21(a) and (b), respectively. Caused by the insufficient supervision of old knowledge, the brightness of diagonal blocks for these classes, especially for the base, gradually decays as the

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Fig. 22. Confusion matrices of our method tested on the last session of three FSCIL datasets with one-way five-shot setting. The statistical results of benchmarks tested repeatedly for T times are adopted and normalized for distinct comparison. (a) Confusion matrices of our method tested on the MSTAR-FSCIL dataset. (b) Confusion matrices of our method tested on the SAR-AIRcraft-1.0-FSCIL dataset. (c) Confusion matrices of our method tested on the Self-FSCIL dataset.

session increases, suggesting an inevitably catastrophic forgetting issue of our method for old knowledge. Besides, the color difference between diagonal and nondiagonal blocks for new classes is salient, demonstrating the powerful capability of our method for identifying new categories once being updated from few-shot samples. Additionally, the color trends and distributions of the two matrices are similar, implicitly suggesting our method's powerful adaptability to learn new knowledge continually and rapidly in various SAR ATR environments.

b) *Overall Matrices:* The matrices of all compared methods tested on the final session (all targets) of the MSTAR-FSCIL, the SAR-AIRcraft-1.0-FSCIL, and the Self-FSCIL datasets are given by Fig 22(a)–(c), respectively. Some conclusions can be provided as follows. Overall, targets of base categories (see the first four diagonal blocks of all matrices for an example) can be accurately classified by the compared benchmarks, benefiting from rich and discerning information provided by abundant training samples. Notably, the matrices of our method tested on the three datasets are clearer and brighter in diagonal blocks than others, indicating a substantial tradeoff between the model's stability for the old knowledge and the plasticity for the new. Besides, most targets are labeled as old categories raised by inaccurate and insufficient information from few-shot new samples. Although the oracle achieves the upper bounded results, the performance gap for recognizing base and novel classes is pronounced, resulting from the long-tailed distributions of samples of different classes. Although solutions for the IL and the FSCIL tasks consider the forgetting issue, their ability to capture discerning features of new classes is still weak due to rare and confused information of few-shot samples, leading to biased predictions for the base and incremental categories.

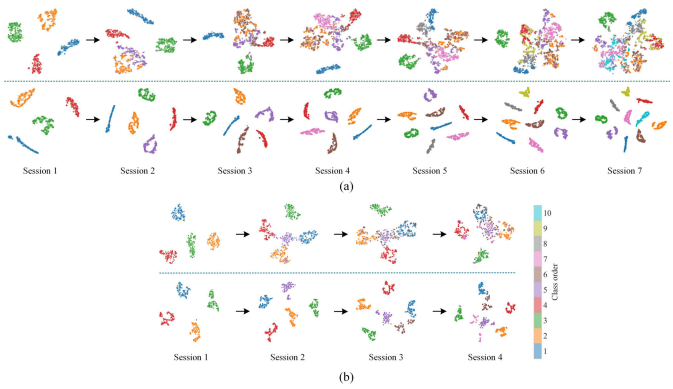


Fig. 23. tSNE of evolved class features at original feature spaces (above the dotted line) and those projected by class subspaces (under the dotted line) at the Grassmannian manifold. The color bar shows categories arranged in order. (a) Results on the MSTAR-FSCIL dataset. (b) Results on the SAR-AIRcraft-1.0-FSCIL dataset.

3) *tSNE of Feature Evolution:* The tSNE of evolved class features at the original and the Grassmannian manifold are investigated in this part. In particular, the results on both the MSTAR-FSCIL and the SAR-AIRcraft-1.0-FSCIL datasets are provided in Fig. 23(a) and (b), respectively. It can be seen that class features projected on the Grassmannian manifold by class-related subspaces are more compact than those at the original, no matter what categories the targets are (vehicle or aircraft) or the SAR imaging platforms (airborne or spaceborne) they are captured by. Besides, even though serious overlaps exist among class features at original spaces after a long-term learning procedure (see the tSNE at the last session for an example), they are still well-separated at the manifold after being transformed by their subspaces, verifying the effectiveness of subspace classifiers on the Grassmannian manifold for target identification.

V. More Discussion

In this section, the effects of hyperparameters, including exemplar budget, loss coefficient, and class margin, are further discussed on the MSTAR-FSCIL dataset, considering its authority and completeness.

A. Exemplar Budget

In our method, the catastrophic forgetting of learned knowledge is resisted in the SLF loss by aligning the model's responses to saved exemplars of old categories during learning. The more class exemplars saved, the richer target information can be utilized for supervision. The performance of our method at the variation of exemplar budgets is given by

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TABLE XII
PERFORMANCE OF OUR METHOD AT DIFFERENT EXEMPLAR BUDGETS

Exemplar budget	Inc.Acc (%) ($t > 1$)			Avg.Acc (%)	PD (%) ↓
	old	new	all		
2	75.12	65.56	73.40	77.19	37.58
3	77.58	65.57	75.60	79.04	35.06
4	79.00	67.16	76.96	80.24	32.84
5	79.96	65.23	77.50	80.69	31.84

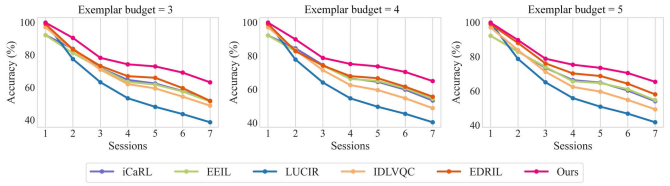


Fig. 24. Performance curves of our method compared with selected exemplar-based benchmarks at different exemplar budgets.

Table XII. It can be seen that our method's performance is consistently improved as the exemplar budget increases. For example, as the number of exemplars grows from 2 to 5, the Avg.Acc increases from 77.19% to 80.69% with a 5.74% improvement on the PD. Furthermore, the same metrics used for judging the classification accuracy of old, new, and all categories at all incremental sessions are also provided as the Inc.Acc (old/new/all). The Inc.Acc old is consistently improved as the numbers of exemplar budgets increase as expected, benefiting from abundant target cues provided by the diversified exemplars.

Since the learned knowledge is preserved and distilled based on saved exemplars, our method's performance curves compared with other exemplar-based benchmarks are plotted in Fig. 24. It can be seen that our method's curves (the red ones) are stably higher than others by a large margin for all exemplar budgets, verifying our method's advantages of learning and discriminating classes based on limited samples.

B. Loss Coefficient

In contrast to learning closed-world classifiers by a single-discrimination loss, dynamically balancing the model's plasticity and stability is crucial for tackling the FSCIL of SAR ATR. In the AASC, the coefficient γ in $\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{SMS}} + \mathcal{L}_{\text{DSS}} + \gamma\mathcal{L}_{\text{SLF}}$ of Algorithm 1 is used to adaptively balance the contributions of knowledge preservation ($\mathcal{L}_{\text{SLF}}$) and adaption ($\mathcal{L}_{\text{SMS}}$,$\mathcal{L}_{\text{DSS}}$). Specifically, three ways to update the γ are exploited. We first set the γ to 1.0, which makes the two contradictory parts contribute equally at incremental sessions. Second, the value is dynamically updated by $((|\mathcal{C}^{\text{old}}|/|\mathcal{C}^t|))^{1/2}$ [47]. Third, we remove the square root and set the value to $(|\mathcal{C}^{\text{old}}|/|\mathcal{C}^t|)$. We mark the three types of γ as the const, slow, and fast based on their session trends shown in Fig. 25(a). The larger the γ, the stronger the model's ability to preserve old knowledge.

Fig. 25(b) shows our method's performance on old and new classes at each session. Thanks to the appropriate balance of the slow γ, the higher performance on old and new classes is achieved by our method at most incremental sessions than the method with the const γ. Caused by a severe bias to old knowledge preservation, our method's ability to learn new concepts is weak and restricted, leading to the most suboptimal performance in identifying new classes. Correspondingly, the performance in classifying all seen classes is given by Fig. 25(c). Our method achieves the highest performance among all the comparisons at each learning session, suggesting the slow γ for optimally balancing the plasticity and stability.

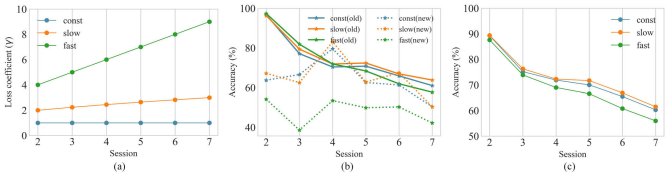


Fig. 25. Performance curves of our method at the variation of loss coefficients. (a) Value of γ at different sessions. (b) Accuracy of different types of classes. (c) Accuracy of all seen categories.

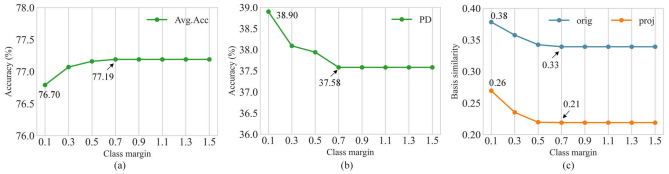


Fig. 26. Performance curves of our method at different $\hat{m}$ of the SMS loss. (a) Avg.Acc curve. (b) PD curve. (c) Basis similarity curves at both the original and the projected spaces.

C. Class Margin

The margins between new and old classes are controlled by the $\hat{m}$ in the SMS loss. The larger the value, the stronger the supervision of enlarging interclass separations. In Fig. 26(a) and (b), the curve on the Avg.Acc increases steadily with a continual decrease in the PD as $\hat{m}$ increases from 0.1 to 0.7. Then, the performance reaches a saturation point when $\hat{m}$ is higher than 0.7. Furthermore, curves of basis similarity between new and old class features at the original and the projected spaces are given by Fig. 26(c). The curves are with similar trends to that of the PD. Therefore, we empirically set the class margin ($\hat{m}$) to 0.7 for optimal learning.

VI. CONCLUSION

We propose an AASC on the Grassmannian manifold to power the DL-based SAR ATR systems with the FSCIL ability to survive in dynamically noncooperative scenarios.

Considering the prominent azimuth dependence and part sparsity of targets in SAR images, three losses (SMS, DSS, and SLF) covering both semantic and manifold levels and an AES strategy are designed to mitigate the stability and plasticity dilemma of the FSCIL and domain challenges of the SAR ATR. For the model's plasticity, the pairwise semantic margins of new classes to the old are enlarged by the SMS. The DSS derived from the Grassmannian geometry maintains the subspace orthogonality. For the model's stability, both spatial and global target structures are aligned in the SLF loss for old knowledge preservation. The AES strategy is designed for selecting representative and complementary exemplars. In experiments, three FSCIL of SAR ATR datasets derived from the MSTAR, the SAR-AIRcraft-1.0, and the self-collected

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images, covering various categories, imaging platforms, and observing situations, are constructed for evaluation. Ablation studies verified the effectiveness of the subspace and the proposed losses and strategies. Extensive experiments conducted on the datasets show the topmost performance achieved by our method compared with various task-specific benchmarks, verifying the effectiveness of the AASC for FSCIL of SAR ATR.

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Yan Zhao (Graduate Student Member, IEEE) received the B.S. degree in information countermeasure technology from the Beijing Institute of Technology, Beijing, China, in 2018, and the M.S. degree in information and communication engineering from the National University of Defense Technology, Changsha, China, in 2020, where he is currently pursuing the Ph.D. degree.

His research interests include incremental learning and few-shot learning in synthetic aperture radar automatic target recognition.



Lingjun Zhao received the B.S., M.S., and Ph.D. degrees in information and communication engineering from the National University of Defense Technology (NUDT), Changsha, China, in 2003, 2004, and 2009, respectively.

She is currently an Associate Professor with the College of Electronic Science and Technology, NUDT. Her research interests include remote sensing information processing and synthetic aperture radar automatic target recognition.



Siqian Zhang received the B.S. degree in electronic information engineering from Hubei University, Wuhan, China, in 2009, and the M.S. degree in photogrammetry and remote sensing and the Ph.D. degree in information and communication engineering from the National University of Defense Technology, Changsha, China, in 2011 and 2016, respectively.

Her research interests include radar signal and image processing, synthetic aperture radar imaging, target characteristic analysis, deep learning, and sparse representation.



Kefeng Ji (Member, IEEE) received the B.S. degree in aerospace engineering from Northwestern Polytechnical University, Xi'an, China, in 1996, and the M.S. and Ph.D. degrees in information and telecommunication engineering from the National University of Defense Technology (NUDT), Changsha, China, in 1999 and 2003, respectively.

In 2003, he joined the College of Electronic Science, NUDT, where he is currently a Professor. He has authored or coauthored over 80 articles. His research interests include signal processing, machine learning, pattern recognition, remote sensing information processing, synthetic aperture radar image interpretation, target detection, recognition, feature extraction, and marine surveillance.

Dr. Ji is a reviewer of several international journals and conferences, such as the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING, the IEEE GEOSCIENCE AND REMOTE SENSING LETTERS, IEEE ACCESS, the IEEE TRANSACTIONS ON INDUSTRIAL INFORMATICS, the *International Journal of Remote Sensing*, *Remote Sensing Letters*, the *European Journal of Remote Sensing*, and *Remote Sensing*.



Gangyao Kuang received the B.S. and M.S. degrees in geophysics from the Central South University of Technology, Changsha, China, in 1988 and 1991, respectively, and the Ph.D. degree in communication and information from the National University of Defense Technology (NUDT), Changsha, in 1995.

He is currently a Professor with the College of Electronic Science and Technology, NUDT. His research interests include remote sensing, synthetic aperture radar (SAR) image processing, change detection, SAR ground moving target indication, and classification with polarimetric SAR images.



Li Liu (Senior Member, IEEE) received the Ph.D. degree in information and communication engineering from the National University of Defense Technology (NUDT), Changsha, China, in 2012.

During her Ph.D. study, she was a Visiting Student with the University of Waterloo, Waterloo, ON, Canada, from 2008 to 2010, for more than two years. From 2015 to 2016, she was a visiting researcher with the Multimedia Laboratory, The Chinese University of Hong Kong, Hong Kong, for ten months. From 2016 to 2018, she was a Senior Researcher with the Machine Vision Group, University of Oulu, Oulu, Finland. She is currently a Full Professor with the College of Electronic Science and Technology, NUDT. Her research interests include computer vision, pattern recognition, and machine learning.

Dr. Liu was the Co-Chair of nine international workshops at Conference on Computer Vision and Pattern Recognition (CVPR), International Conference on Computer Vision (ICCV), and European Conference on Computer Vision (ECCV). She served as the Leading Guest Editor for special issues in the IEEE TRANSACTIONS ON PATTERN ANALYSIS AND MACHINE INTELLIGENCE (IEEE TPAMI) and the *International Journal of Computer Vision*. She is serving as the Leading Guest Editor for the IEEE TPAMI special issue on Learning with Fewer Labels in Computer Vision. She currently serves as an Associate Editor for the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, the IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS FOR VIDEO TECHNOLOGY, and *Pattern Recognition*. Her papers have currently over 10 500 citations according to Google Scholar.

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